



Allied Engineering & Services (Pvt.) Ltd.

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02 x CATERPILLAR CG-132B-12, 600 eKW / 750 KVA
Continuous RATED, 0.8 PF, 415V, 50 Hz



picture may differ from actual genset

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Prepared for

ASTRA Engineering & Construction
LLC

Techno-Commercial Proposal

AESL-GTC-60808 REV00

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Ex-factory Shipment Prior to:

31 – July – 2025

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1. ABOUT US

Among Caterpillar's dealership network globally **Allied Engineering & Services (Pvt.) Ltd.** enjoys the sole dealership title of Caterpillar Inc. within the region with which they have successfully capitalized on the existing market, in Pakistan.

Headquartered in Karachi, Allied Engineering & Services (Pvt.) Ltd., has offices in geographically strategic locations within the region i.e.: Lahore, Islamabad, Multan, Faisalabad, Peshawar, and Quetta. The highly trained, capable and qualified engineers are available 24/7 to provide the required support, both during and after the warranty period.

2. PRODUCT OFFERING

2.1 Continuous Power

- **02 Nos.** Caterpillar Gas Gen set model **CG132B-12** of **750 KVA / 600 eKW**, W/O Fan, With Electric Driven Pumps, Continuous Power, at **45°C, 70% Relative Humidity, 15m**, 1500 RPM, 50 HZ, 415 V, 0.8 P.F.
- Net Rating after parasitic loading at site condition is 540ekW, 1500 RPM, 50 HZ, 415 V, 0.8 P.F.

When Ambient temperature at site is > 45degC, Adiabatic (Mist) cooling in remote radiator will be activated automatically to minimize deration with minimum water consumption.

Continuous Power Definition:

It is the output available without varying load for an unlimited time. Continuous power is in accordance with ISO8528, AS2789 and BS5514.

1.1 Technical data of the genset

Design conditions		
Intake air temperature / air humidity	°C / %	45 / 70
Installation height	m	14
NOx raw emissions (without SCR)	mg/Nm ³ @ 5% O ₂	240
NOx raw emissions (with SCR)	mg/Nm ³ @ 5% O ₂	100
Combustion gas data(*)		
Combustion gas	Associated gas	
Methane number		77
Lower heat value	kWh/Nm ³	10.43

Gas density	kg/Nm³	0.87
Genset		
Engine	CG132B-12	
Configuration Code	P	
Generator	Generator - Leroy-Somer - LSA 49.3 L9 or similar	
Voltage	V	415
/voltage range	/ %	
Frequency	Hz	50
Operation mode		Grid-parallel and island operation
Energy balance		
Electrical terminal power at cos φ = 0.800 / ind./overexcited	kW	596
Cooling water heat (± 8%)	kW	385
NT mixture cooler heat (± 8%)	kW	38
Fuel use (+ 5%)	kW	1438
Electrical efficiency	%	41.4

2.2 Fuel Consumption

1	Site Air Intake Temp	degC	45
2	Relative Humidity		70%
3	Natural Gas GCV	Btu/Sft3	1,259.27
4	Natural Gas LHV	Btu/Sft3	1007.41
		MJ/Nm3	37.55
		kWh/Nm3	10.43

CG132B-12 @ 415V 0.8 PF								
S.No.	Performance Parameter	Unit	Load Factor					Applicable Tolerance on stated values for Performance
			100%	95%	90%	75%	50%	
1	Electrical Output @0.8 p.f	ekW	596	566	536	447	298	Fixed power mode
2	Energy Input	kWt	1438	1371	1304	1103	767	+5% Tolerance
3	Specific Fuel Consumption	kWt/kWe	2.41	2.42	2.43	2.47	2.57	+5% Tolerance
4	Fuel Consumption (LHV) basis	BTU/kWh	8232	8261	8294	8417	8786	+5% Tolerance
5	Fuel Consumption (GCV) basis	BTU/kWh	9096	9129	9164	9300	9708	+5% Tolerance
6	Fuel Consumption	Nm3/ekWh	0.231	0.232	0.233	0.237	0.247	+5% Tolerance
7	Fuel Consumption	Sm3/ekWh	0.244	0.245	0.246	0.250	0.261	+5% Tolerance

Guaranteed fuel consumption at 596 EKW Continuous power shall be **0.244 Nm³ / eKW - hr @ 1259 Btu / ft³ GCV, 77 Methane Number** & is subject to + 5% ISO tolerance.
The fuel consumption will be displayed on the Generator Control Panel

2.3 Delivery

The ex-factory delivery of the Gen sets is approx. 36-42 weeks after the establishment of irrevocable L/C and as per General Terms & Conditions mentioned in proposal.

2.4 Package Commissioning at site

Day rates shall be quoted later at the time of negotiations based on scope between Buyer and Seller.

2.5 Warranty

The warranty is based solely on our current General Terms and Conditions. It applies only to defects, which occur within 24 months from the initial commissioning or 36 months from the date of delivery,

2.6 Price Validity

All the prices are valid till one month from quote and thereafter subject to confirmation by AESL. **Imported and local prices are exclusive of taxes, custom duties and GST etc.** For payment, terms please refer to attached terms and conditions for imported and local.

The technical and commercial proposal cover the equipment selection and pricing based on the interpretation of the information supplied to Allied Engineering. The proposal is complete herein. Any change, deviation or additions to the scope or bill of materials not specified in the technical and commercial bids will require suitable adjustment in the price, and may require additional engineering and / or delayed delivery.

3. SELLER'S SCOPE OF SUPPLY

3.1 Overview

Number	Title
Genset	
Engine Parts	
2	Engine configuration
2	Design and efficiency package
2	Basic parts
2	Exhaust gas system, Turbo components, with wastegate
2	Exhaust turbo charger
2	Cooling system
2	Torsional vibration damper
2	Torsional vibration damper protection
2	Fly wheel
2	Fly wheel casing
2	Starter equipment
2	Engine Mounting
2	Ignition device
2	Lube oil system
2	Gas mixer
2	Air filter system
2	TPEM-Control Unit
2	Primer and Topcoat Engine - Primer
2	Conservation
Engine Test	
2	Engine acceptance test with customer participation
Standard Tool Set	
1	Tool Kit large scope
Tools Set for Maintenance	
1	Tool kit service Barring device
Tool Set for Repairs	
1	Tool kit repair standard
Spare Parts for New Engines	
2	Spare parts for new engines Low / 12000 h
Genset Parts	
2	Genset definition
2	Generator - Leroy-Somer - LSA 49.3 L9
2	Base frame with integrated oil tank
2	Elastic genset mounting
2	Elastic coupling
2	Fasteners
2	Crankcase ventilation
2	Attachment fresh oil Pump 230 / 400 / 460 V/50 / 60 Hz
2	Flexible ventilation pipe
2	Primer and Topcoat for Genset - Cat-yellow EATY008
2	Repair Paint - Cat-yellow EATY008
Pre-Lubrication	
2	Pre Lubrication Genset - 230 / 400 / 460 V
Additional Parts for Lubrication System	
2	Solenoid valves 230V for the automatic refilling of the built-in fresh oil tank
Name Plates for Complete Engine	
2	Labeling of the genset - English (US)
Packing for Complete Genset	

2	Seaworthy packing for genset: wooden pallet and foil cover
Fuel System	
Inlet Fuel Pressure Regulator	
2	Pre pressure gas train Dungs DN 25/25/50 Outlet 150.00 mbar
Zero-Pressure Gas Control Train	
Gas Train	
2	Gas train DN 50/50 -
Connecting Accessories for Zero Pressure Train	
2	Elastic connection line DN 50
2	Connection Accessories for Gas Train DN 50 - Steel
2	Inlet Manifold DN 40/50
Heating System	
Mixture cooling circuit	
Radiator	
4	Rubber bellow joint - DN 150 / 16 PN
4	Set of flanges for rubber bellow joint - 150 DN
Cooling Water Pump	
2	Cooling water pump 32 - 400 V/50 Hz - 2 kW
Three-Way-Valve	
2	Three-way valve - DN 32 / PN 16 - 24 V - 5 kN
2	Counter Flange Set DN 32 - Set of 3
Sensors	
2	Temperature Sensor - 120 mm
2	Safety Valve - bar 3.00 - G 1 1/4 x G1 1/2 - directly heated
2	Water low level switch (Type 933.2)
2	Ventilation valve
2	Differential pressure switch
Shut Off Valve	
2	Shut-off valve DN 32
2	Counter flange set DN 32 - Steel
Expansion Vessel	
2	Expansion vessel - 100 Ltr. - PN 6
2	Lock-off valve for expansion tank
2	Expansion tank manometer
2	Cut-off valve for Manometer (test port)
Engine Cooling Circuit	
Radiator	
4	Rubber bellow joint - DN 150 / 16 PN
4	Set of flanges for rubber bellow joint - 150 DN
Cooling Water Pump	
2	Cooling water pump 50 - 400 V/50 Hz - 8 kW
2	Counter Flange Set DN 50 - Set of 2
Three-Way-Valve	
2	Three-way valve - DN 65 / PN 16 - 24 V - 5 kN
2	Counter Flange Set DN 65 - Set of 3
Sensors and Accessories	
2	Temperature Sensor - 120 mm
2	Safety Valve - bar 3.00 - G 2 x G 2 1/2
2	Water low level switch (Type 933.2)
2	Ventilation valve
2	Differential pressure switch
2	Gas and dirt separator DN 65
2	Counter Flange Set DN 65 - Set of 2
Electrical Preheating	
2	Electric Preheating - 400 / 480 V/6 kW/with flanges

2	Counter Flange Set DN 65 - Set of 2
Throttle Valve	
2	Throttle Valve - DN 50/ PN 16
2	Counter flange set DN 50 - Steel
Shut Off Valve	
2	Shut-off valve DN 80
2	Counter flange set DN 80 - Steel
Expansion Vessel	
2	Expansion vessel - 400 Ltr. - PN 6
2	Expansion tank manometer
2	Lock-off valve for expansion tank
2	Cut-off valve for Manometer (test port)
Exhaust Gas System	
Exhaust Gas Silencer	
2	Exhaust gas silencer 300 DN - heat resistant steel - axial/axial/horizontal - 60 dBA
4	Exhaust gas counter flange DN 300 - complete - Steel
Sensors	
2	Thermocouple NiCr-Ni exhaust gas
2	Thermocouple NiCr-Ni after catalyst
2	Exhaust back pressure sensor
Lubricant System	
Lube Oil Tank	
1	Fresh oil tank - single walled - 1450 Ltr.
Lube Oil Pump Fresh Oil	
2	Lube oil pump, fresh oil
2	Hose pipe (between tank and pump) connections 1" / 1/2"
2	Overflow Valve
Sensors and Accessories	
2	Signal tower for oil
2	Overfill protection for fresh oil tank
Starting System	
Batteries	
2	Starter battery in single storage box
Engine Room Installation	
Ventilation	
Sensors	
2	Temperature sensor circulation air - 160 mm
2	Temperature sensor ambient air - 160 mm
Electrical System	
Genset Management (TPEM)	
2	TPEM-Basic System
1	TPEM Preconfiguration RAM - Intermediate subscription
1	TPEM Preconfiguration for engineering data collection
1	Island Mode - Functionality
1	TPEM Token Set Level 50
2	Cabling 25m IEC
2	TPEM I/O Controller
Switchgear	
Auxiliary Panel	
Cabinet	
2	Basic Scope
Power Plant Components	
2	SCR Catalyst Supply and control, 400V, IEC
Enclosure/Engine Room Accessories	
2	Electrical system

Common Control Panel	
Basic Components	
1	Basic scope of supply
Modbus	
1	Modbus RTU 2
Low-voltage	
Power Panel (GCB)	
2	Generator Circuit Breaker Panel, 400 V, 1600 A, 3-pole
Services	
Packing	
Plant Components	
1	Packing plant components
Transport	
1	Transportation includes the following Transport from Germany port to Khi port Transport from Khi to Allied facility for packaging of gensets inside enclosure Transport from Allied facility to Khi port for re-export to Abu Dhabi Transport from Khi port to Abu Dhabi port
Service Packages	
Engineering Package	
1	Engineering Package DTO
PM Package	
1	PM Package DTO High
Documentation	
Documentation First language	
2	Documentation first language - standard documentation
Documentation second language	
2	Documentation second language - standard documentation
Documentation third language	
2	Documentation third language - standard documentation
Certificates and Labels	
2	EG Declaration for Machinery Directive
Pre- Pressure Regulator	
2	Pre- Pressure Regulator to knockdown gas pressure from 17.5 bar to 1.75bar
Selective Catalytic reduction	
2	Selective catalytic reduction (SCR) for maintaining Nox in limit
Remote radiator	
2	Radiator set for Jacket water and after cooler adiabatic cooling
Enclosure	
2	Sound attenuated enclosure 85dBA @ 1m with Fire safety protection alarms

4. PRICING SUMMARY

S.NO	DESCRIPTION	QTY	PRICE USD (CNF KARCHI BASIS)
1	Caterpillar Gen Set Model CG132B-12 of 600 ekW / 750kVA Prime Power, 1500 RPM, 50 HZ, 0.8 PF, 11000 V (with complete supply scope) @Site Conditions: 45degC Temp, 70% Relative Humidity	02 Nos	\$ 1,350,000

4.1 Balance of Plant

S.NO	DESCRIPTION	QTY	CNF Price \$
	BALANCE OF PLANT as per ADN-AEC-ME-SPC-026 MR Gas Genset	2 Lot	\$ 490,000
1	Adiabatic Remote radiator with controls <i>Radiator set for Jacket water and after cooler adiabatic cooling with automatic mist energization to minimize deration</i>	2 Set	Included In Balance of Plant
2	Selective Catalytic reduction Selective catalytic reduction (SCR) for maintaining Nox and particulate matters in limit	2 sets.	Included In Balance of Plant
3	Sound attenuated Enclosure	2 sets.	Included In Balance of Plant

	Sound attenuated enclosure 85dBA @ 1m with Fire safety protection alarms , Filters , Fans, AC , Lighting and Shutters as per enclosure drawing attached		
4	Pre- Pressure Regulator Pre pressure regulator for for site gas pressure with gas flowmeter	2 sets.	Included In Balance of Plant
5	Tests TVA, Hydraulic testing, insulation resistance and SAT	2 Jobs.	Included In Balance of Plant
6	Earthing Earthing for Package equipment	2 Jobs.	Included In Balance of Plant

PROJECT FINANCIAL SUMMARY

Sr#	AESL Seller Scope Description	CNF Dubai Price
1	CG132B-12 Genset	
1.1	2 x CG132B-12 @ 750kVA 415V, 50Hz)	\$ 1,350,000
2	Balance of Plant	
2.1	Complete supply of scope as above (section 4.1)	\$ 490,000
	Total	\$ 1,840,000

5. GENERAL TERMS AND CONDITIONS

PRICE:	All prices are on CFR Abu Dhabi basis.
VALIDITY:	All prices are valid till one month from quote date. Any validity extension beyond this date or change in specs is subject to our review and confirmation, which may result in revision of prices and delivery date. Your acceptance is recognized upon the receipt of workable letter of Credit (according to the terms below) by the beneficiary .
TERMS OF PAYMENT:	Letter of Credit – irrevocable, confirmed and workable Letter of Credit payable at sight to be established in favor of beneficiary as under: DYNAMIC ENGINEERING SERVICES AND SUPPLIES PTE LTD Company Registration No 201726184D 105 CECIL STREET NO. 20-04, THE OCTAGON SINGAPORE (069534) <i>20% advance under LC</i> <i>50% advance on presentation of genset shipment status shipped from Germany factory to Karachi port</i> <i>30% balance on packaged enclosure on presentation of shipping documents of re-export to Abu Dhabi</i>
ADVISING BANK:	Workable letter of credit (L/C) must be transmitted through full telex to any Prime European/American bank in Geneva. L/C must be negotiated at the counter of Advising bank in Geneva and L/C must contain the reimbursement clause with Opening bank A/C # of any USA/UK bank for reimbursement purpose.
BANK CHARGES:	All foreign and local bank charges to OPENER'S account.
PARTIAL & TRANS-SHIPMENT:	Partial shipment and trans-shipment should be ALLOWED in the L/C.
PORT OF SHIPMENT/EXIT:	Port of shipment mentioned in the L/C must be “Any EUROPEAN/USA/SINGAPORE Port ”.
COUNTRY OF ORIGIN:	L/C must stipulate Country of Origin as USA, EUROPEAN COMMUNITY or SINGAPORE.
NEGOTIATION PERIOD:	21 days must be allowed for negotiating the L/C from the date of issue of Bill of Lading.
NON-NEGOTIABLE DOCUMENTS DISPATCH PERIOD:	Beneficiary will dispatch One Set of Non-negotiable shipping documents to the customer within 21 days from the date of issue of Bill of Lading.
MARINE INSURANCE:	Marine Insurance should be carried out by the Buyer. Complete Postal Address of the Insurance Company is required in the L/C. Fax Number or P.O. Box No. is not sufficient.
COMMERCIAL INVOICE:	“Four Folds” of Commercial / Product Invoice shall be provided.
PACKING LIST:	One Set of Packing List shall be provided.

FORWARDER:	Caterpillar will use their own forwarder and their appointed shipping lines. Caterpillar will not accept any other forwarder and shipping line if specified in the L/C.
ORDER ACCEPTANCE AND SHIPMENT:	Upon the receipt of a workable Letter of Credit by beneficiary , the equipment ordered will be scheduled for ex- factory delivery within 36-42 weeks from LC acknowledgement date with Caterpillar. The ex-factory lead time can be adjusted for earliest availability at the time of order.
CERTIFICATE & COURIER RECEIPT:	Please DO NOT ask for Certificate and Courier receipts as it delays the flow of shipping documents.
FORCE MAJEURE:	This offer is subject to standard force majeure conditions.
DUTY & TAXES:	All customs levies/duties/taxes are excluded from our offer and shall be charged at actual at the time of invoice.
GENERAL:	This offer should be read in conjunction with the quantities and prices mentioned in the original offer of which this forms a part. Any changes in requirement will be subject to our confirmation.
REIMBURSEMENT CLAUSE	Final LC Draft must have following Reimbursement Clause : "IF documents negotiated are in order and strictly comply credit terms THEN negotiating bank is authorize to claim reimbursement from..... (here, opening bank will mention their bank a/c details of any USA bank for reimbursement purpose)..... after two working days of negotiation of original documents. Negotiating bank must inform opening bank by tested telex / swift within 24 hours the details of negotiation and copy of the same must accompany the original documents."
TITLE TRANSFER	In case of Ocean or Air shipment, legal title to beneficial ownership of product sold to customer will transfer to customer once the shipment is unloaded at the port of entry immediately prior to customs frontier of the importing country.
SHIPPER DETAILS	<u>SHIPMENT AND SHIPPER ON B/L</u> Shipment will be made directly by Caterpillar. Shipper on the B/L will be as follows. M/S CATERPILLAR S.A.R.L ROUTE DE FRONTENEX 76 1208 GENEVA - SWITZERLAND Ph: 00 41 22 849 44 44 Fax: 00 41 22 849 45 44

6. TECHNICAL SPECIFICATIONS OF CG132-12 415V VOLTS

1.2 Genset

Gas engine and generator housings are coupled fixed and mounted on a common base frame for vibration-proof installation through spring elements. The drive train engine / generator is connected torsionally flexible via a highly elastic, axially pluggable flange coupling.

1.2.1 Engine

Industrial engine for stationary application.

Four-stroke Otto engine with lean-mixture charging, cylinders in V arrangement with water cooling for cylinders and mixture.

Counterclockwise rotation direction according to DIN 6265 (seen on flywheel end).

Main engine components

Crankcase

One-piece crankcase made of gray cast iron with screwed-on oil tray.

Driving gear

Forged crankshaft stored suspended in housing, flywheel flanged and with gear ring for the starter, spur gear unit to drive the camshaft, torsional vibration damper on the opposite end of the flywheel

Piston with piston rings

Forged con-rod

Water-cooled cylinder liners, can be replaced individually, with specialized honing for low lube oil consumption.

Cylinder head

Individual cylinder heads with four valves and seat rings, spark plug positioned in center of combustion chamber, O-ring oil seal at the valve shafts

Valve drive using tappets and push rods.

Oil lubrication

Pressure lubrication via a gear pump, lube oil heat exchanger in the engine cooling circuit, oil filter in the main stream.

Charging

Exhaust turbocharger for compressing the gas-air mixture, 2-stage mixture cooler for recooling (low temperature level in a separate cooling circuit).

Starter

24 V electric starter.

Fuel supply / Mixture treatment

Multi-gas mixer for dosage of the gas-air mixture

Dry air filter

Throttle organ with electric actuator for the speed and power control

Electronic control and monitoring of the air-gas mixture at the mixer-actuator via combustion chamber temperature (1 temperature sensor per cylinder).

Sensors

- Sensors for:
- Temperatures
 - o Cooling water
 - o Intake air
 - o Mixture
 - o Oil
 - o Combustion chamber
- Pressure sensor
 - o Oil
 - o Crankcase
- Speed
- Gas mixer position recording
- Knock sensors
- Oil level

Most sensors, sometimes installed multiple times, are wired to a central wiring harness.

Ignition system

Non-wearing, high-voltage ignition system controlled by microprocessors with low-voltage distribution, one ignition coil per cylinder.

Knock monitoring and control

Anti-knock control with knock sensor monitoring (one sensor per two cylinders) for operation free from knocking with best possible performance and efficiency under observance of the emission limit values.

Connections

Connection to the plant's lines for vibration decoupling

HT circuit: Rubber expansion joints with counterflange

LT circuit: Flexible stainless steel expansion joints

Exhaust outlet: Stainless steel expansion joint with counterflange

Fuel gas inlet: see fuel systems, if applicable

Boost pressure control

Boost pressure control and adjustment to various ambient conditions via a bypass valve (exhaust wastegate) in the exhaust flow

Corrosion protection

Corrosion protection of the engine or genset for protection during transportation and any subsequent storage at the place of destination; maximum length of protection of 24 months after delivery, when stored in closed dry room.

Engine Test with Customer Participation

The engine is tested at our test benches using the test bench's generator and the measurements are recorded.

All test bench equipment for recording of measurements is calibrated.

A control system that belongs to the test bench is used for the test. A genset test with customer's control system is not possible.

The measurements are carried out in a standardized testing procedure at 100% standard testing power.

The recorded measurement data are summarized in the engine performance report, that is part of our documentation.

Conditions for 50Hz operation: Mains parallel mode, no island mode, no load steps.

The following measured values are provided:

Power, speed, lubricating oil pressure, gas consumption at methane number 80, efficiency, intake air temperature, exhaust gas temperature after turbine, receiver pressure, O₂ dry, exhaust gas emission NO_x (Reference: 5% O₂ in dry exhaust gas).

Spare parts package for new systems

The spare parts package for the new system essentially contains wear materials which are required due to the selected running time 12000 and the maintenance schedule for the gas quality Low specified in the project. The spare parts package always refers to the standard scope of supply for the engine. Special components such as water separators for IFE drainage are not included. Gasket sets for turbochargers are also not included, as every turbocharger has an individual gasket set. Primarily included are materials such as prechamber spark plugs, air and oil filters and various gaskets. The spare parts packages are adapted to the current developments, meaning they correspond to the current version when delivered.

1.2.2 Three-phase Internal-pole Synchronous Generator

Type: Leroy-Somer LSA 49.3 L9 or similar (*)

Voltage: 415 V

Frequency: 50 Hz

Three-phase synchronous generator, salient-pole machine, brushless, self-excited, with damper cage, star point dissolvable, with insulation suitable for tropical and humid ambient conditions (max. 90 % humidity).

General structure

The generator comprises the main generator as internal-pole machine, an exciter as external-pole machine and a voltage controller with a power output stage, the power unit of which is supplied by an auxiliary winding.

The power of the exciter stator is supplied by the voltage controller. The three-phase alternating voltage which is induced in the exciter rotor is rectified by the rotating diode bridge and is then supplied to the field winding of the main generator. The voltage stabilization of the main generator under variable load in island mode and the setting of a specified power factor or the specified reactive power in mains parallel mode is performed by changing the field current via the power actuator in the voltage controller.

Protection type and ventilation

Type of ventilation IC 01 acc. to IEC 60034-6 open cooling circuit with self-ventilation and protection class IP23 acc. to IEC 60034-5.

Stator

The insulation of the stator winding corresponds to insulation class H/ Temperature rise according to generator data sheet according to IEC 60034-1.

Rotor

The insulation of the rotor and field winding corresponds to insulation class H/ Temperature rise according to generator data sheet according to IEC 60034-1.

The rotor is balanced with half key in compliance with quality G 2.5 according to ISO 21940-1. Max. overspeed (spin speed) according to IEC 60034-1.

Winding

To reduce the harmonics, the generator is equipped with a 2/3 pitch winding. The windings are insulated with a high-quality cast resin by using vacuum-pressure impregnation (VPI) which protects the windings from moisture, dust, ash, and abrasive particles.

Shaft bearings

By default the generator is designed as a double-bearing machine. Roller bearings suitable for relubrication.

Voltage controller

Digital voltage controller with diode failure monitoring, mounted in the auxiliary terminal box.

Voltage nominal value setting $\pm 10.00\%$ U_N .

Standards and specifications

IEC 60034-1, ISO 8528-3, NEMA MG 1-32

Standard components / assembly groups

- Central terminal box with main terminals and star point, auxiliary terminal box with terminal strip and accessories
- Generator controller with functions for regulating the voltage, the reactive power or the shift factor, incl. the voltage and current transformers required for regulation
- Temperature monitoring with 6 PT 100 – sensors in the winding (3 x active, 3 x spare)
- an auxiliary winding
- Anti-condensation heating: 230 V $\pm 5\%$, 1 x 250 W
- Current transformer: 1200/1/1 A
 - o Core 1: 10 VA, 0.5FS5 (measurement, not calibrated)
 - o Core 2: 10 VA, 5P10 (general protection)
 - o
- With anti-magnetic cable outlet plate without cable flange and without glands.

Electrical data and properties

- Voltage accuracy, static ± 0.5 to 1 % of the rated voltage
- Speed change + 3% / - 5 % of the rated speed, cold and warm machine
- Total harmonic distortion factor THDU (Total Harmonic Distortion) of the voltage (Ph-Ph) at no-load ≤ 4 %
- Unbalanced load $I_{\text{Inverse}} / I_N \leq 8$ % according to IEC 60034-1
- Radio interference suppression level Class A, Group 1 according to DIN EN 55011; VDE 0875-11
- Sustained short-circuit current with a 3-pole terminal short-circuit: minimum threefold rated current for 5 s
- Generator overload capacity 10 % of rated power for 1 h within 6 h

*CES reserves the right to change the alternator supplier and type during offer period. The genset data may thereby change slightly. The power output will not change. CES will confirm the alternator type, brand and alternator data sheet with the order confirmation.

1.2.3 Genset Parts

Base frame

Stable steel construction for supporting the engine and generator. Rubber spring elements are mounted between the engine-generator compound and base frame for vibration decoupling. The main side beams contain integrated tanks, on the one hand to increase the engine oil volume and on the other as a fresh oil tank for refilling.

Coupling

Torsion-elastic, axial plug-in flange coupling for connection of engine and generator. The rubber element can be radially removed.

Flange genset coupling bell housing

The engine and generator are connected to each other in a torsionally resistant manner by a flywheel housing and coupling bell housing.

Oil level control

Automatic lube oil refilling, control using an immersion probe for oil level monitoring (min./max.) and for controlling the solenoid valves and the fresh oil pump.

Prelubrication pump

Electrical prelubrication pump on genset; the engine and integrated lube oil tank can be emptied for oil change by switching a three-way valve.

Fresh oil pump

The consumed engine oil is refilled by an electric pump at the genset.

Paintwork

According to industrial standards, paintwork of the assembled genset consists of:

- Primer, dry film thickness 40^{+30} μm
- Topcoat in the shade Cat-yellow EATY008 dry film layer 90^{+40} μm
- 1 can refinishing paint, loose in the shade of the topcoat.

Additional parts for lubrication system

Additional components are delivered "loose" and are not mounted on the genset.

The following components are delivered with the aggregate:

- 2 solenoid valves 230 V with connection parts for lube oil inlet
- 1 set of flexible hoses for lube oil inlet and outlet connections

Labeling

Language on the labeling of the genset: English (US)

Seaworthy packaging on wooden skids with film

Packaging, consisting of a cover with bubble wrap, desiccant and barrier layer made of film and screw connection of the genset on wooden skids. Suitable for sea transport in a container. Loose parts in transport boxes

1.3 System components

The components stated in the following chapters are included in the scope of supply and are delivered "loose". Assembly and any cabling is performed on site by the client, as long as no installation for these components is listed.

The stated technical data may change at order.

1.3.1 Fuel system

Package pressure gas control line

The purpose of the pre-pressure control line is to reduce the gas pressure of the upstream supply network at the customer handover point to the input operating pressure of the zero pressure control line of the gas engine.

Gas type:	According gas composition
Nominal width (outlet side):	DN 50
Gas pre-pressure min:	1 bar
Gas pre-pressure max:	1.8 bar

Gas control line as a tested functional assembly made from individual components with EC prototype test certificates in accordance with the EC pressure equipment directive and the EC gas equipment directive, suitable for gases of the gas families 1, 2 and 3 (in the gaseous phase) in accordance with DVGW G 260.

Provided it complies with the regulations of the applicable manufacturer's declaration, the gas control line is deemed to be "permanently technically sealed" in accordance with TRBS 2152 Part 2, Section 2.4.3, and can thus

be excluded as the cause of an explosive atmosphere.

- 1 ball valve
- 1 manometer
- 1 gas filter
- 1 pre-pressure controller
- 1 safety shut-off valve (SAV) with end position switch for the message "triggered"
- 1 safety blow-off valve (SBV) with local gas leak display
- 1 ball valve for manual emptying and venting
- 1 outlet molding for connection of the regulator and SAV measurement lines
- 2 counterflanges for installation of the gas control line in the gas line

The equipment of the gas control line may vary depending on the gas or operation type. The bill of materials separately created at order is binding.

Vent, blow-off and relief lines that vent into the open air are to be provided on-site.

Zero pressure gas control line

Gas type:	According gas composition
Nominal width (outlet side):	DN 50
Manufacturer:	Dungs

Gas control line as a tested functional assembly made from individual components with EC prototype test certificates in accordance with the EC pressure equipment directive and the EC gas equipment directive, suitable for gases of the gas families 1, 2 and 3 (in the gaseous phase) in accordance with DVGW G 260.

Provided it complies with the regulations of the applicable manufacturer's declaration, the gas control line is deemed to be "permanently technically sealed" in accordance with TRBS 2152 Part 2, Section 2.4.3, and can thus be excluded as the cause of an explosive atmosphere.

- 1 ball valve
- 1 gas filter
- 1 manometer
- 1 pressure monitor for minimum gas pressure
- 2 automatic shut-off valves, valve class A, group 2
- 1 flame arrester, suitable for selected fuel gas

- 1 thermal monitoring with trigger isolated switch amplifier
- 1 zero pressure controller
- 1 outlet molding in the case of external measuring point of the zero pressure controller
- 2 counterflanges for installation of the gas control line in the gas line

The gas control line is designed for an inlet pressure of **150 mbar** with a permissible accuracy of AC 10 with a pressure fluctuation frequency < 10/ h. To secure the control accuracy of the outlet pressure, the maximum permissible inlet pressure is 200 mbar. The outlet pressure (OPd) is 0 mbar +/-3 mbar.

The equipment of the gas train may vary depending on the gas or operation type. The bill of materials separately created at order is binding.

If applicable, a secure ventilation and outlet line that leads to outdoors is to be provided on-site.

Elastic connection line

Elastic connection line with steel mesh guard for flexible connection of the gas line to the gas-air mixer.

Nominal width: DN 50

Gas manifold

Pipe elbow (90° bend for assembly on the gas-air mixer) to match the expansion joint.

1.3.2 Mixture cooling circuit

Cooling water pump

Constant-speed, single-stage, dry-rotor inline centrifugal pump for light industrial applications, for installation in pipelines, with maintenance-free floating seal and hydraulically-optimized impeller.

Discharge height*	18 m
Cooling water volumetric flow	9.00 m ³ /h
Connection flanges	DN 32 / PN 16
Rated engine output	2 kW
Voltage	400 V
Energy efficiency class	IE3
Glycol content	33 %

* The pump has a provisional design and is subject to the Detail Engineering department, a pressure loss of 0.4 bar for on-site piping of the cooling circuit has been taken into account.

3-way valve

Three-way control valve with actuator, optimized as mixing valve (can be used as a diverting valve)

Housing made of EN-GJL-250, 3-way form, spindle made of 1.4021. Electric thrust drive, 5 kN, degree of protection IP 65, mechanical stroke indicator, manual emergency adjustment.

Connection flanges DN 32 / PN 16

Motor voltage/frequency 24 V

Temperature sensor after radiator

Temperature sensor (PT 100 with transmitter) with separate threaded protective sleeve made of stainless steel, cover on the connection head with clip fastening, neck pipe made of stainless steel.

Operating voltage 8 to 35 VDC

Measuring range 0 to 200 °C

Outlet signal 4 to 20 mA

Installation length 120 mm

Safety valve

Diaphragm safety valve with upstream sealing gasket that is separated from the diaphragm.

Max. heat output, directly heated 395 kW

Opening overpressure: 3.00 bar

Low water safety device

Water level limiter with glass float and with magnetic transfer of the float movement to a microswitch.

Max. operating overpressure 10 bar

Max. operating temperature 120 °C

Installation position main axis vertical

Vent valve

Float vent in pressed brass design, with ready sealed shut-off device and corresponding shut-off valve.

Maximum temperature 120 °C

Maximum pressure 10 bar

Differential pressure monitor for flow monitoring

Differential pressure switching device , with microswitch.

max. operating pressure 16 bar

pressure connection (cutting ring screw joint) 12 mm

measuring range 0-1 bar

Shut-off valve

Ring valve with engaging lever, clamping fitting with centering lugs for assembly between two flanges.

Connection flanges: DN 32 / PN 16

Expansion tank

Membrane pressure expansion tank, built according to DIN EN 13831, certification in accordance with EU Pressure Equipment Directive 2014/68/EU, exterior coated, membranes cannot be replaced.

Nominal volume* Ltr. 100

* During the design phase, an average pipe length of approx. 50 m and a pre-pressure of 1.5 bar was assumed.

1.3.3 Engine cooling circuit

Cooling water pump

Constant-speed, single-stage, dry-rotor inline centrifugal pump for light industrial applications, for installation in pipelines, with maintenance-free floating seal and hydraulically-optimized impeller.

Discharge height* 33 m

Cooling water volumetric flow 37 m³/h

Connection flanges DN 50 / PN 16

Rated engine output 8 kW

Voltage 400 V

Energy efficiency class IE3

Glycol content 33 %

* The pump has a provisional design and is subject to the Detail Engineering department. A pressure loss of 0.2 bar (CHP) or 0.5 bar (power module, or CHP with integrated emergency cooling circuit without pump) for on-site piping of the cooling circuit has been taken into account.

3-way valve

Three-way control valve with actuator, optimized as mixing valve (can be used as a diverting valve)

Housing made of EN-GJL-250, 3-way form, spindle made of 1.4021. Electric thrust drive, 5 kN, degree of protection IP 65, mechanical stroke indicator, manual emergency adjustment.

Connection flanges DN 65 / PN 16

Motor voltage/frequency 24 V

Temperature sensor after radiator

Temperature sensor (PT 100 with transmitter) with separate threaded protective sleeve made of stainless steel, cover on the connection head with clip fastening, neck pipe made of stainless steel.

Operating voltage 8 to 35 VDC

Measuring range 0 to 200 °C

Outlet signal 4 to 20 mA

Installation length 120 mm

Safety valve

Diaphragm safety valve with upstream sealing gasket that is separated from the diaphragm.

Max. heat output, directly heated 1017 kW

Opening overpressure: 3.00 bar

Low water safety device

Water level limiter with glass float and with magnetic transfer of the float movement to a microswitch.

Max. operating overpressure 10 bar

Max. operating temperature 120 °C

Installation position main axis vertical

Vent valve

Float vent in pressed brass design, with ready sealed shut-off device and corresponding shut-off valve.

Maximum temperature 120 °C

Maximum pressure 10 bar

Differential pressure monitor for flow monitoring

Differential pressure switching device , with microswitch.

max. operating pressure 16 bar

pressure connection (cutting ring screw joint) 12 mm

measuring range 0-1 bar

Gas and dirt separator

The gas and dirt separator removes the smallest of micro-bubbles and dirt particles from the cooling system. This improves the coolant quality, meaning that plant components have a longer service life and engine problems are avoided.

Connection: Flange connection DN 65

max. volumetric flow rate: 49.00 m³/h

Pressure loss at max. volumetric flow rate: 0.1 bar

Electric preheating for cooling water

Electric preheating to preheat and maintain engine heat when stationary. Configured as a tank, equipped with heating elements, for installation in the engine cooling water circuit in accordance with the P&I diagram.

Heating output 6 kW

Voltage 400 / 480 V

Design: with flanges

Certificate:

Throttle valve

Ring valve with worm gear transmission and handwheel for infinitely variable setting of the throttle function, clamping fitting with centering lugs for assembly between two flanges.

Connection flanges: DN 50 / PN 16

Shut-off valve

Ring valve with engaging lever, clamping fitting with centering lugs for assembly between two flanges.

Connection flanges: DN 80 / PN 16

Expansion tank

Membrane pressure expansion tank, built according to DIN EN 13831, certification in accordance with EU Pressure Equipment Directive 2014/68/EU, exterior coated, membranes cannot be replaced.

Nominal volume* Ltr. 400

* During the design phase, an average pipe length of approx. 50 m and a pre-pressure of 1.5 bar was assumed.

1.3.4 Exhaust system**Exhaust silencer**

Exhaust silencer configured as a combined reflection-absorption silencer.

Material, silencer	heat resistant steel
Pressure loss	15.00 mbar
Sound pressure level at a distance of 10 m	60 dBA
Exhaust gas connection flanges	300 DN / Hole pattern as per PN10
Exhaust inlet nozzle	axial
Exhaust outlet nozzle	axial
Installation position	horizontal

Accessories

Set of counterflanges, seals and screws for connection of the Exhaust silencer into the exhaust line

Nominal size:	DN 300
Counterflanges:	complete
Material:	Steel

Exhaust gas temperature sensor

Thermocouple (NiCr-Ni with transmitter) for monitoring the exhaust gas temperature, cover on the connection head with clip fastening, neck pipe made of stainless steel.

Operating voltage 8 to 35 VDC

Output (impressed current)	4 to 20 mA
Installation length	350 mm

Temperature sensor downstream of catalytic converter

Thermocouple (NiCr-Ni with transmitter) for monitoring the exhaust gas temperature, cover on the connection head with clip fastening, neck pipe made of stainless steel.

Operating voltage	10 to 35 VDC
Measuring range	0 to 1000 °C
Output (impressed current)	4 to 20 mA
Installation length	350 mm

Exhaust back-pressure monitor

Pressure sensor for monitoring the maximum permissible exhaust gas back pressure downstream of the engine, for installation in the exhaust line downstream of the engine (turbocharger) and upstream of the catalytic converter or exhaust silencer. Diaphragm pressure monitor with microswitch, scope of supply including approx. 400 mm protective tube.

Switching range	mbar	5 to 500
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1.3.5 Lube oil system

Fresh oil tank, single-walled

Single-walled fresh oil tank made from steel plate on bracket design, with general building inspection approval certificate, internally flushed out with corrosion protection oil, externally 2-K paintwork in RAL 7032.

Content	ca. 1450 Ltr.
Length	1500 mm
Width	760 mm
Height	1680 mm

- Level indicator consisting of two level switches (reed contacts) that are height-adjustable.
- Local contents display configured as a mechanical fill level meter with gear drive. With continuous display (indicator) in percent of the total height.

Lube oil pump, fresh oil

Gear pump with three-phase motor, coupling, intermediate flange, pressure limiting valve and terminal box.

Discharge rate approx. (SAE 30, 80°C) 0.60 m³/h

Voltage	230 / 400 / 460 V
Motor output	0 kW

Signal tower for oil, indoor configuration

Signal tower with auditory alarm for "lube oil tank full", wall-mounted

Operating voltage: 24 V

Degree of protection: IP66

Volume: 92 - 102 dB(A)

1.3.6 Starting System

Starter batteries in protective case

Lead starter batteries (dry pre-charged, filling required on site) are used to start the genset. Delivery in acid-proof battery protective case made from plastic and the associated battery connection terminals (without cabling to the starter)

Number of protective cases	2	Unit
Batteries per protective case	1	Unit
Output voltage	24	V
Battery capacity	143	Ah

1.3.7 Engine room system

Temperature sensor for recirculating air

Temperature sensor (PT 100 with transmitter), cover on the connection head with clip fastening.

Operating voltage	8 to 35 VCD
Output (impressed current)	4 to 20 mA
Installation length	160 mm

Temperature sensor for ambient air

Temperature sensor (PT 100 with transmitter), cover on the connection head with clip fastening.

Operating voltage	8 to 35 VCD
Output (impressed current)	4 to 20 mA
Installation length	160 mm

1.4 Electrical system

1.4.1 Control System TPEM (Total Plant & Energy Management)

The Total Plant & Energy Management (TPEM) system assumes the regulation and control functions for the operation of a gas genset made by Caterpillar Energy Solutions GmbH. The plant control as well as functions for the synchronization of the genset. The plant control as well as functions for the synchronization of the generator circuit breaker, mains circuit breaker (if necessary) and generator protection are integrated in the TPEM. Operation can be either local or via remote access. Thanks to its monitoring functions, the TPEM protects the genset and the plant against impermissible operating conditions and enables a long service life and economic efficiency.

Components

The system comprises the following components:

- TPEM Control Unit
- TPEM Control Cabinet
- TPEM I/O Controller
- Access to TPEM Rendezvous Server
- TPEM Remote Client

The TPEM Control Unit is mounted on the genset and controls the engine. It is connected via the integrated TPEM Connection Box to the TPEM Control Cabinet.

The TPEM Control Cabinet is the central control cabinet. It contains a 15" Touchscreen (HMI), the TPEM Multi Function Relay (MFR), the TPEM Remote Plant Gateway (RPG), interfaces to external controls and the TÜV tested TPEM safety chain.

- The HMI serves for local visualization and operation of the plant. Available in 27 different languages, it is intuitive and user-friendly in design and contains a central operating element for all regulation, control, service and monitoring functions of the plant, an operating log, histories as well as further diagnostic and service functions (including auxiliary genset test mode).
- The MFR controls the generator circuit breaker and mains circuit breaker (if required) and must be connected to this at the plant end. The TPEM system supports the following generator and mains protection functions in accordance with the ANSI standard:
 - Generator protection function:

- ANSI# 81O/U Overfrequency and underfrequency
- ANSI# 59/27 Overvoltage and undervoltage
- ANSI# 51V Time-dependent overcurrent
- ANSI# 50 Present overcurrent
- ANSI# 46/47 Unbalanced load
- ANSI# 32/32 R Overload and reverse power
- ANSI# 25 Synchronization check
- Decoupling protection on the generating unit:
 - ANSI# 81O/U Overfrequency and underfrequency
 - ANSI# 59/27 Overvoltage and undervoltage
 - ANSI# 78 Time-dependent overcurrent
 - ANSI# 25 Synchronization check
- The RPG is a preconfigured router. Via the customer-end Internet connection it autonomously establishes a VPN connection to the TPEM Rendezvous Server in the data center of the Caterpillar Energy Solutions GmbH. A standard Internet connection at the plant site is required for this.
- The connection to external controls is supported via Modbus RTU or TCP.

Assembly and cabling of the TPEM Control Cabinet are carried out on site by customer, as long as assembly for these components is not indicated in this quote.

The TPEM I/O-Controller is the interface between the TPEM system and the auxiliary drives of the plant. It prepares all inputs and outputs necessary to safely regulate and control the auxiliary drives belonging to the genset. Feedback can be imported into an external control when required.

The TPEM I/O Controller is supplied loose for installation in the auxiliary drive cabinet (MCC). If the auxiliary drive cabinet (MCC) and its commissioning are not included in the scope of supply, the customer is responsible for performing the electrical integration, check and commissioning of the TPEM I/O Controller.

The TPEM Rendezvous Server allows the TPEM Remote Client Software included in the scope of supply to connect to the control system via remote access

The access rights depend on the particular user (Tokenlevel). They extend from monitoring via remote diagnostics and maintenance through to control and regulation of the plant.

Functions and features (extract)

The TPEM system supports the following functions and features, amongst others:

- Automatic program sequence for starting and stopping the gas genset
- Integrated digital engine speed and power control
- TÜV-approved security chain with all required connections in the TPEM Control Cabinet
- Monitoring and regulation of cooling, heating and lubricating oil circuits
- Test mode for verification of connected actuators and auxiliaries
- Electronic operating log with detailed records of warning, fault and operating messages

A complete list plus details of the individual functions and features can be found in the TPEM operating manual.

Certification

With respect to unit certification of the model series, all relevant electrical characteristics of the generation unit were measured and disclosed in the unit certificate in the course of type testing. This also includes the entire TPEM control.

Technical data TPEM Control Cabinet

Free-standing cabinet dimensions (LxWxH)	600 x 800 x 2000 (mm)
Base dimensions (LxWxH)	600 x 800 x (2x 100) (mm)
Color	RAL 7035
Operating temperature	0 - 40°C
Power supply	24V DC
Cable entry	Top or bottom
Protection class	NEMA 1 / IP20
Standards	optional IEC 61439-2; UL 508a; CSA 22.2

TPEM Pre-configuration RAM - Intermediate Subscription

TPEM is pre-configured to allow collection of customer data and this option will initiate the activation process to make genset data and digital features available via the connectivity platform Remote Asset Monitoring (RAM). This will not lead to extra cost until a subscription contract is made separately between Service partner and end customer. The required hardware (PLE network manager) is part of the basic scope of supply of the TPEM CC. The customer data collection transmits and stores genset and BoP data whenever the genset is online. Data volume depends on selected subscription models and can range from a few MB to 30MB/day.

Any connection cost has to be borne by the customer.

Features of RAM intermediate:

Unlimited number of users

Events & Alarms

Data sampling rate 1/min

Real time alarms and configurable alerting capabilities via text or email

Multiple viewing options, for single assets or an entire fleet

Configurable dashboards to track critical KPI's

Operational data available for download

16 Engine Parameters

29 Electrical Parameters

RAM AssetMonitor mobile App

An FAQ document explaining RAM and how to get the genset connected as well as the required Data Governance consent form can be found in the attachments.

TPEM Pre-configuration for engineering data collection

TPEM is pre-configured to allow collection of engineering data to transmit genset and BOP data to an engineering data server. The data can be used to analyze system operation and/or to troubleshoot specific genset issues.

The aim is to generate insights for improving operation and development of next generation gensets, data is shared on a voluntary basis without any obligation for CES of direct rewards. The customer will not have a direct interface to the data but can benefit through feature development which can be made available via the CatConnect program or through genset modifications in hard- or software.

The engineering data collection transmits and stores genset and BOP data whenever the genset is online. Data volume depends on application and can range up to 300MB/day.

Any connection cost has to be borne by the customer. In case of bandwidth conflicts, the bandwidth is shared between the engineering and customer data collection channels.

TPEM Token Set Level 50

The TPEM token set level 50 contains a TPEM USB storage and a TPEM USB token with authorization level 50. These can be used for data transfer (TPEM USB storage) or for authentication to the TPEM system (TPEM USB token). Suitable for use on the USB port of the TPEM CC or on your own computer when using the TPEM remote client.

MCC Control and monitoring

- Installation and cabling of the TPEM IO Controllers, for control commands and actuation of the auxiliary drives
- Auxiliary drive cabinet feeder 400 V / 230 V [3Ph, 4W, PE], approx. 250 A via manual circuit breaker
- Actuator for leakage check of gas control line or safety shut-off valve (SAV) or electric vent valve (includes one alternative per plant)
- Actuator of the voltage adjustment in generator

- Battery charger for engine starter battery 24V / 20A I / U regulated, primary-clocked version with battery voltage monitor
- Voltage sensor for soft starter (if soft starter is required)
- Control for engine cooling water pump 400 V, max. 22 kW with soft start
- Control for mixture cooling pump 400 V, approx. 15 kW
- Supply of prelubrication pump 400 V, approx. 2 kW
- Control for cooling water preheating 400 V, 2 x 3 kW
- Control for generator stationary heating 230 V, approx. 1 kW
- Control for three-way valve mixture cooler 24V DC
- Control for lube oil solenoid valves
- Installation of type-tested switching device for detonation monitoring of the gas control line (only for biogas plants)
- Supply of the oil pump 400 V, approx. 1 kW

Voltage supply 24 V for auxiliary drive and TPEM cabinet

- Fuse distribution 24 V control circuits with circuit breakers
- Fused outgoing circuit with auxiliary circuit 24V / 35A for TPEM (battery-buffered)
- Fused outgoing circuit (B 6A) for protecting the 24 V supply of the TPEM IO Controllers
- Battery charger 24V, 40A I / U regulated, primary-clocked version with battery voltage monitor for control battery
- Control battery (gel version) 24V, 20Ah, maintenance-free and non-fumigating, installed in switch cabinet

SCR KAT Supply and control

- 2x voltage supply 230 V
- Potential-free contact "Genset on/off"
- Analog genset power output (4-20mA)

Electrical system

- 2x Residual current operated circuit breaker 4 pole 40/0,03A
- 3x Automatic circuit breaker 1 pole, B 16A for Schuko socket 230 V
- 2x Automatic circuit breaker 1 pole, B 10A for lighting 230 V
- 1x Automatic circuit breaker 1 pole, B 16A for air conditioning unit 230 V
- 4x Automatic circuit breaker 1 pole, B 16A for space heating 230 V

1.4.2 Common control panel

Master plant control basic equipment

Switch cabinet

A Programmable Logic Controller SPS type Siemens Simatic S7 is installed into the GCP. It contains an overvoltage protection 24V DC and a 6" operating device Siemens Touchpanel in the door front. The plant functions are parameterized on this touch panel.

The PLC is programmed with the following basic software:

- Selection of the gas gensets via contacts of the customer's central control technology during power peaks (peak load operation) or heat demand (heat-led operation)
- Heat-led operation depending on the heating water return temperature using PT100 heat sensors, selection and deselection of gas gensets according to heat requirements, including selection of maximum 2 heating boilers with additional heat requirements
- Data coupling from TPEM to Simatic S7 SPS via TTY interface
- Data coupling from generator multi-functional device to Simatic S7 SPS via Profibus DP or CAN bus interface
- Display of operating and fault messages of all gas gensets on the operating panel (English)
- Operating hours compensation of the modules (start change program) for the same service life
- Additional display and recording of maximum 16 common fault messages (digital inputs) of all modules

Example of configuration (determined specific to each project):

- Fault overvoltage / undervoltage triggered
- Fault overfrequency / underfrequency triggered
- Mains fault / vector surge triggered
- section switch open / fault
- Fault cabin ventilation
- Fault fresh oil pump
- Fresh oil tank empty
- Fresh oil tank overfilled
- Waste oil container full
- Waste oil container overfilled

Communication interface Modbus RTU

Transmission of all measured values und fault messages of the gas genset and generator from the TEM, the multifunction relay and the MPCC to the customer's control technology via the interface (includes a hardware interface). The interface is designed as a slave.

7. BUYER'S SCOPE OF SUPPLY

Civil Works	• Power house building, Control Room Building, trench cover, drainage and sludge disposal, Lighting system, general exhaust, control room air conditioning and overhead water storage tank. • Storage location for equipment at site • Site offices, accommodation, and meal of designated AESL staff. • Construction of Foundation Pads, Electrical and Mechanical Trenches
Equipment Supply & Installation	<u>Mechanical:</u> ❖ Fire alarm and firefighting equipment (Outside enclosure) ❖ Sludge pump / Trench water drain pump ❖ Steps at Catwalk around equipment ❖ Bulk Storage Tank with piping valves and fittings. ❖ Fuel Transfer pumps ❖ Plant complete ventilation system. ❖ Hot water heat recovery exchangers, Hot water and Exhaust fired absorption Chillers, associated 3 way valves, exhaust dampers and its controls for cogen plant <u>Electrical:</u> ❖ Incoming switch gear(s) and load take-off panels ❖ Power cable, Auxiliary cable and Control Cable ❖ Transformers. ❖ ATS, Bus Coupler, Tie Breaker Panels, outgoing breakers and its related controls for parallel operation. ❖ Solar Synch Module if applicable. ❖ Bus Coupler Module if applicable. ❖ Site Acceptance Test Load Bank & Fuel. ❖ Auxiliary standby power supply for motor control center ❖ Woodward compatible 0-3VDC load share lines on existing gensets ❖ Lightning protection system ❖ Actual load for testing and commissioning of individual genset and system testing as per operation philosophy. ❖ Hook up with existing distribution system (if any) ❖ Auxiliary Supply 24 / 110 VDC & 230/400 VAC ❖ Lube Oil bulk reservoir tank ❖ Remote monitoring of Power plant including SCADA system ❖ Plant network studies for protection coordination.
Consumables	❖ Portable water, fresh water etc for cooling system refills. ❖ Commissioning consumables including lube oil, coolant, battery electrolyte.

Others	❖ Marine insurance of imported items from port of exit to port of entry. ❖ Transportation and insurance, custom clearance of generator & other imported material from Abu Dhabi port to site including lifting, handling equipment like crane, fork lifter etc. ❖ Plant commissioning to be negotiated. ❖ Local insurance of complete erection and installation works ❖ Operation and maintenance of plant as per manufactures guidelines. ❖ Electrical power required at site for installation activities ❖ Safe storage area for keeping installation material at site ❖ All the required Tag plates. ❖ Fuel and load bank for genset testing ❖ Any kind of legal inspection / approval required (from boiler inspector, explosive inspection and electrical inspection etc. or any other.)
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8. ATTACHMENTS

- TECHNICAL DATASHEET
- SUPPORTING DOCS

60876R1 CG132-12 @ 600eKW (<45degC)



Technical data

596 kWel; 415 V, 50 Hz; Acc. to gas analysis

Design conditions

Inlet air temperature / rel. Humidity:	[°C] / [%]	45 / 70
Altitude:	[m]	14
Exhaust temp. after heat exchanger:	[°C]	150
NO _x raw emissions genset (tolerance -8 %):	[mg/Nm ³ @5%O ₂]	240

Fuel gas data: ²⁾

Methane number:	[-]	77
Lower calorific value:	[kWh/Nm ³]	10,41
Gas density:	[kg/Nm ³]	0,87
Acc. to gas analysis		
Analysis CO ₂	[Vol%]	5,39
N ₂	[Vol%]	0,25
O ₂	[Vol%]	0,00
H ₂	[Vol%]	0,00
CO	[Vol%]	0,00
CH ₄	[Vol%]	83,62
C ₂ H ₄	[Vol%]	0,00
C ₂ H ₆	[Vol%]	8,81
C ₃ H ₆	[Vol%]	0,00
C ₃ H ₈	[Vol%]	1,69
C ₄ H ₈	[Vol%]	0,00
C ₄ H ₁₀	[Vol%]	0,17
C ₅ H ₁₂	[Vol%]	0,02
C _x H _y	[Vol%]	0,00
H ₂ S	[Vol%]	0,06
H ₂ O	[Vol%]	0

Genset:

Engine / Configuration code:	CG132B-12	P
Speed / Mean piston speed:	[1/min] / [m/s]	1500 / 8
Configuration / number of cylinders:	[-]	V / 12
Bore / Stroke / Displacement:	[mm]/[mm]/[dm ³]	132 / 160 / 26
Compression ratio:	[-]	13
Mean effective pressure:	[bar]	18,9
Mean lube oil consumption at full load:	[g/kWh]	0,1
Generator:	Leroy Somer LSA 49.3 L9 or similar (*)	
Voltage / voltage range / cos Phi:	[V] / [%] / [-]	415 / 10 / 0,80
Speed / frequency:	[1/min] / [Hz]	1500 / 50

*CES reserves the right to change the alternator supplier and type during offer period. The genset data may thereby change slightly.
The power output will not change. CES will confirm the alternator type, brand and alternator data sheet with the order confirmation.

Energy balance

Load:	[%]	100	75	50
Electrical power COP acc. ISO 8528-1:	[kW]	596	447	298
Engine jacket water heat:	[kW ±8%]	385	292	209
Intercooler LT heat:	[kW ±8%]	38	27	18
Lube oil heat:	[kW ±8%]			
Exhaust heat with temp. after heat exchanger:	[kW ±8%]	266	219	164
Exhaust temperature:	[°C ±25°C]	403	425	447
Exhaust mass flow wet / dry:	[kg/h]	3572 / 3190	2695 / 2404	1859 / 1656
Combustion mass air flow:	[kg/h]	3451	2603	1794
Radiation heat engine / generator:	[kW ±8%]	22 / 25	20 / 19	18 / 14
Fuel consumption:	[kW+5%]	1439	1105	778
Electrical / thermal efficiency:	[%]	41,4 / 45,3	40,4 / 46,3	38,3 / 48,0
Total efficiency:	[%]	86,7	86,7	86,3

System parameters ¹⁾

Ventilation air flow (comb. air incl.) with ΔT = 15K	[kg/h]	18700
Combustion air temperature minimum / design:	[°C]	10 / 45
Exhaust back pressure from / to:	[mbar]	30 / 50
Exhaust volume flow wet / dry:	[Nm ³ /h]	2696 / 2416
Maximum pressure loss in front of air cleaner:	[mbar]	5
Zero-pressure gas control unit selectable from / to: ²⁾	[mbar]	20 ³⁾ / 200
Pre-pressure gas control unit selectable from / to: ²⁾	[bar]	0,5 / 10
Starter battery 24V, capacity required:	[Ah]	143
Starter motor:	[kWel.] / [VDC]	5,4 / 24
Lube oil content engine & extension / clean oil tank:	[dm ³]	420 / 320*
Dry weight engine / genset:	[kg]	2766 / 5796

Cooling system

Glycol content engine jacket water / intercooler:	[% Vol.]	33 / 33
Water volume engine jacket / intercooler:	[dm ³]	43 / 5
KVS / Cv value engine jacket water / intercooler:	[m ³ /h]	30 / 14
Jacket water coolant temperature in / out:	[°C]	78 / 88
Intercooler coolant temperature in / out:	[°C]	48 / 52
Engine jacket water flow rate from / to:	[m ³ /h]	22 / 37
Water flow rate engine jacket water / intercooler:	[m ³ /h]	35 / 9
Water pressure loss engine jacket water / intercooler:	[bar]	1,4 / 0,4
Engine jacket water pressure outlet min / max:	[bar rel.]	2 / 2,5

Page 1 / 2

S3331968EC42448

1) See also "Layout of power plants".

2) See also Techn. Circular 0199-99-3017

3) Minimum pressure may be higher, depending on project conditions.

*) optional

60876

Frequency band f [Hz]																	Sound power level L _W [dB(A)]																	L _{WA} [dB(A)]		S [m ²]	
Air-borne noise ⁴⁾																																					
L _{W,Totz} [dB(lin)]																																					
Exhaust noise ⁵⁾																																					
L _{W,Totz} [dB(lin)]																																					

4) DIN EN ISO 9614-2 (s=±4 dB)																	5) Measured in exhaust pipe (f < 250 Hz: ±5 dB; f > 250 Hz: ±3 dB)																	L _W : Sound power level																	S: Area of measurement surface (S ₀ =1m ²)																	6) DIN 45635-11, Appendix A																

4) DIN EN ISO 9614-2 (s=±4 dB)

5) Measured in exhaust pipe (f ≤ 250Hz: ±5dB; f > 250Hz: ±3dB)

L_W: Sound power levelS: Area of measurement surface (S₀=1m²)

6) DIN 45635-11, Appendix A

ALTERNATOR ELECTRICAL DATA

LSA 49.3 L9 / 4P

LS Reference: DS_A493_L9_415V_600ekW

Date: 11/02/2021

V6.04 - 02/2021

Main data:						M	
Power:	750	kVA	600	kWe	626	kWm	1
Voltage:	415	V	Frequency:	50	Hz		1
Rated voltage range:	+10% / -10%		Speed:	1500	rpm		1
Power factor - Lagging:	0,8						1
Nominal current:	1 043	A	Phases	3			
Insulation / Temperature rise:	H / F		Connexion	Star serial			1
Cooling:	IC01		Winding type:	p2/3			1
			Winding:	- 6 Wires			1
Ambient temperature:	40	°C					1
Altitude:	1000	m	Overspeed (rpm)	1800			1
Duty: Base Rating			Total Harmonic Distortion (THD)	< 5%			1

Efficiency (Base 600 kWe)						IEC
	25%	50%	75%	100%	110%	
Power factor - Lagging: 0,8	92,90	95,41	95,90	95,88	95,81	1
Power factor - Lagging: 1	93,29	96,03	96,76	96,99	97,02	1

Reactances (%) - (Base 750 kVA)						
Unitary impedance (1 per unit) = 0,229633 ohms						
		Unsaturated		Saturated		
		Direct axis		Quadrature axis		
Synchronous reactance	Xd	234	152	Xq	119	77
Transient reactance	X'd	14,1	12,0	X'q	119	77
Subtransient reactance	X''d	13,0	11,1	X''q	22,8	19,4
Negative sequence reactance	X2	17,9	15,2			
X0	1,9	Zero sequence reactance		Reactive power limitation		
Xl	6,5	Stator leakage reactance		at Un		-450 kVAR
Xr	7,9	Rotor leakage reactance		at Un + 10%		-450 kVAR
Kc	0.66	Short-circuit ratio		at Un - 10%		-450 kVAR

Time constants (s)						
		Direct axis		Quadrature axis		
Open circuit transient time constant	T'do	1,73		T'qo	NA	1
Short-circuit transient time constant	T'd	0,104		T'q	NA	1
Open circuit subtransient time constant	T''do	0,016		T''qo	0,071	1
Subtransient time constant	T''d	0,015		T''q	0,014	1
Ta	0,036	Armature time constant				1

Resistances (%)						
Ra	1,6	Armature resistance	R0	0,6	Zero sequence resistance	1
X/R		X/R ratio (without unit)	R2	3,6	Negative sequence resistance	

Special Grid Code requirement: leading PF = 0,95 / Voltage: 415V

Rating is provided for the specified temperature rise, by resistance measurement according to IEC60034-1

According to: I.E.C. 60034.1 - 60034.2 - NEMA MG 1-32

Products and materials shown in this catalogue may, at any time, be modified in order to follow the latest technological developments.

#REF!

ALTERNATOR MAIN CURVES

LSA 49.3 L9 / 4P

LS Reference: DS_A493_L9_415V_600ekW

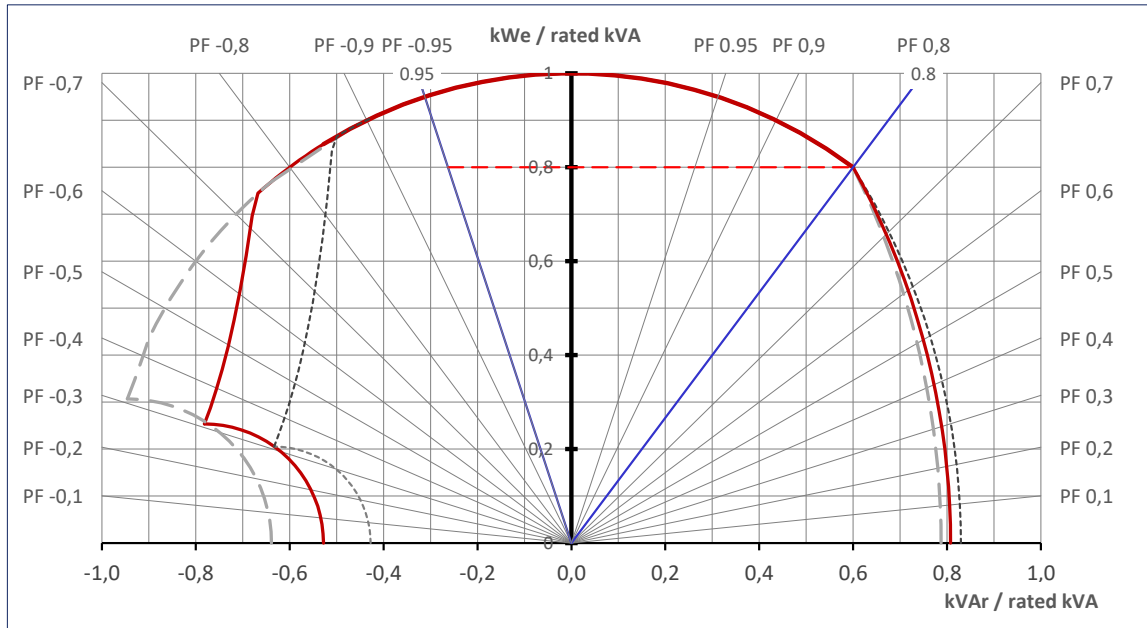
Date: 11/02/2021

750kVA - 415V - 50 Hz

V6.04 - 02/2021

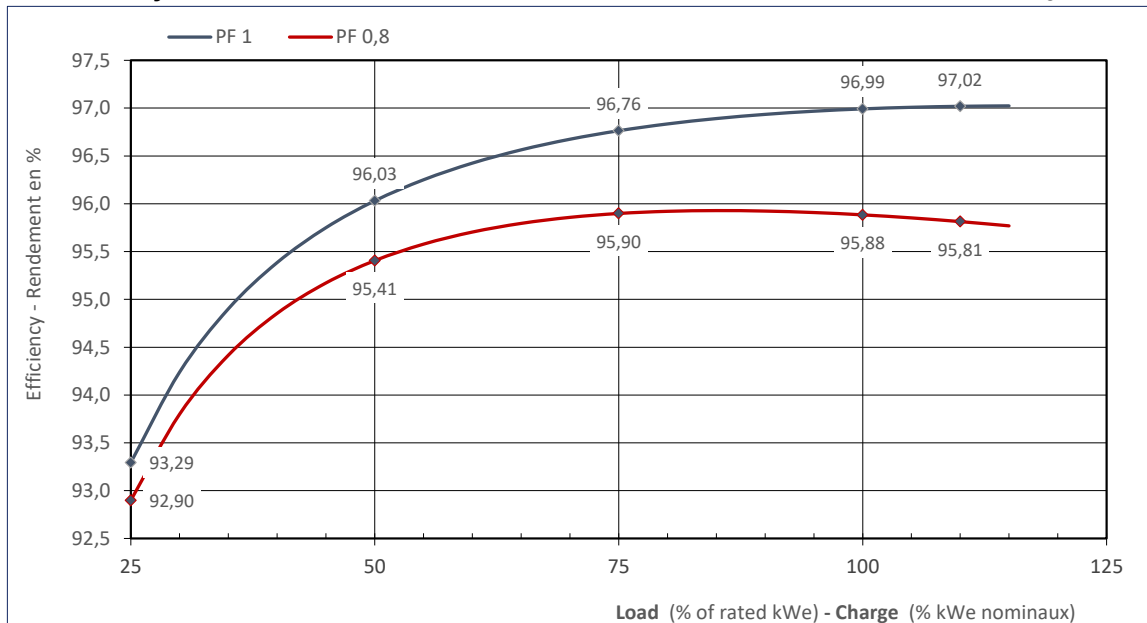
Capability Curve

— — —	Umax	+ 10%	457	V
— — —	Un		415	V
— — —	Umin	- 10%	374	V



Efficiency Curves

According to: IEC



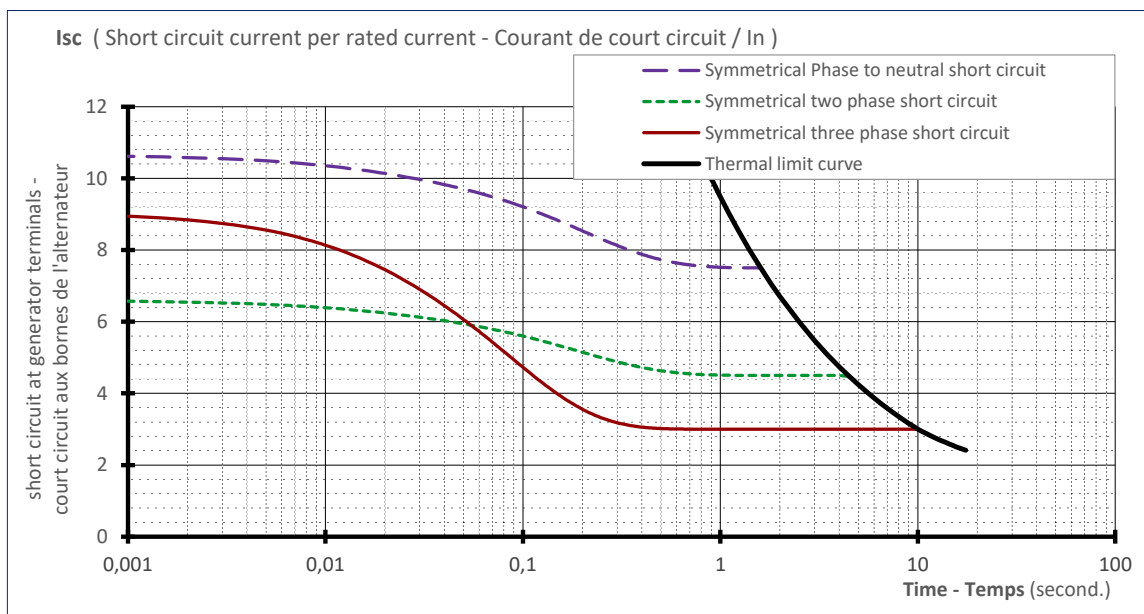
ALTERNATOR MAIN CURVES

LSA 49.3 L9 / 4P

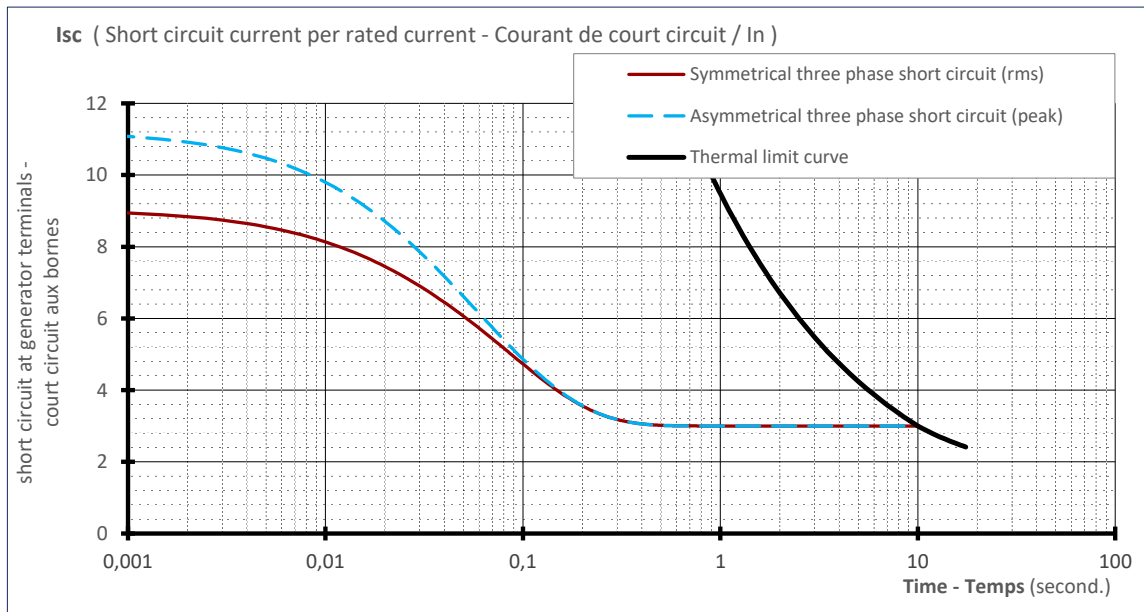
LS Reference: DS_A493_L9_415V_600ekW

Stator Current decrement curves

Symmetrical phase to neutral short-circuit	—	initial	11 073	A	10,6 x In	
Symmetrical two phase short-circuit	- - -	max	6 853	A	6,6 x In	In = 1043 A
Symmetrical three phase short-circuit	—	value	9 329	A	8,9 x In	
Thermal Limit	—					



Asymmetrical three phase short-circuit — IP 11 408 A 10,9 x In

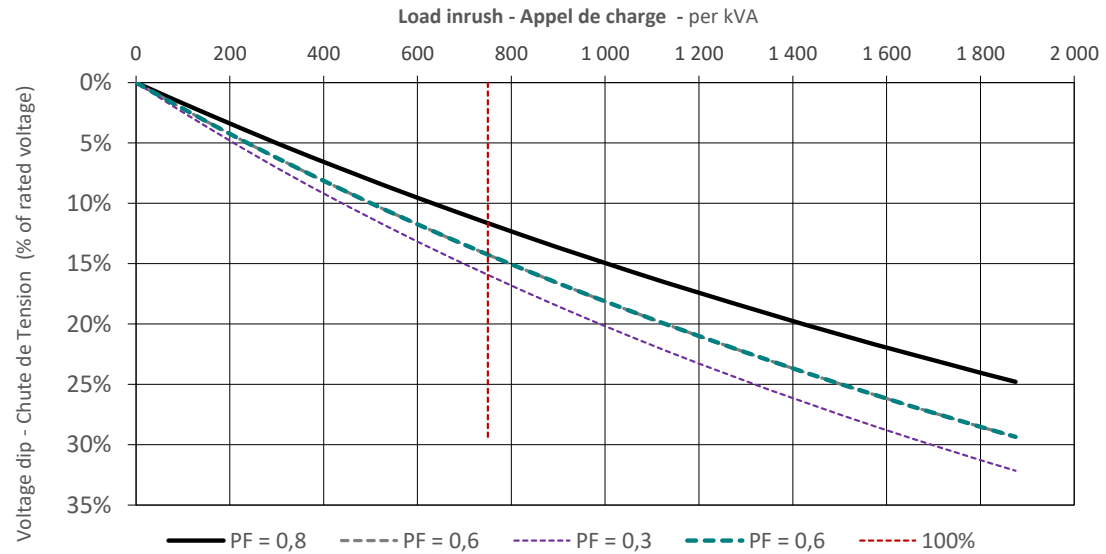


ALTERNATOR MAIN CURVES LSA 49.3 L9 / 4P

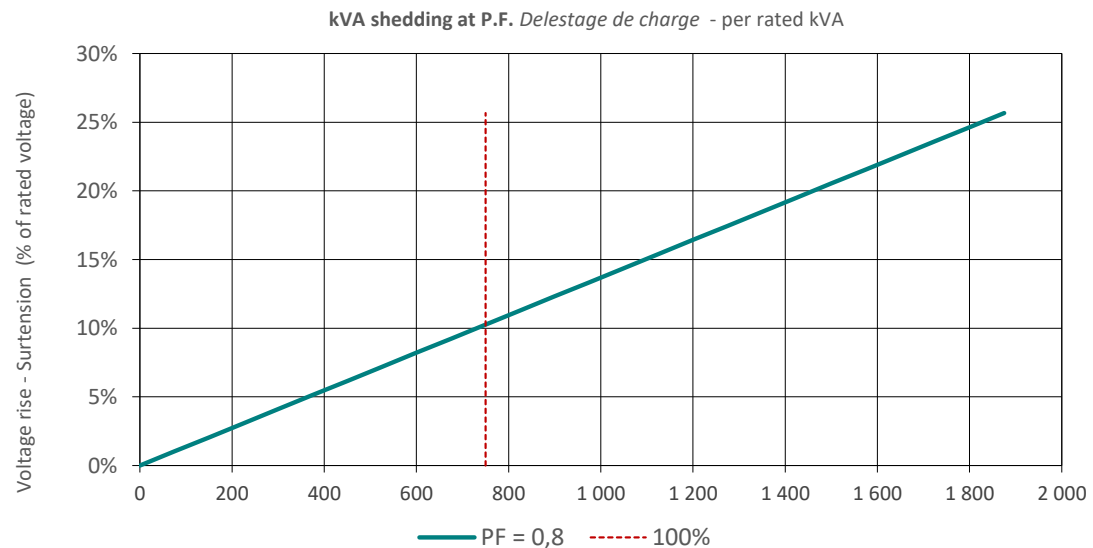
LS Reference: DS_A493_L9_415V_600ekW

Transient Voltage Variation

Transient voltage dip curve versus load impact



Transient voltage rise curve versus load rejection

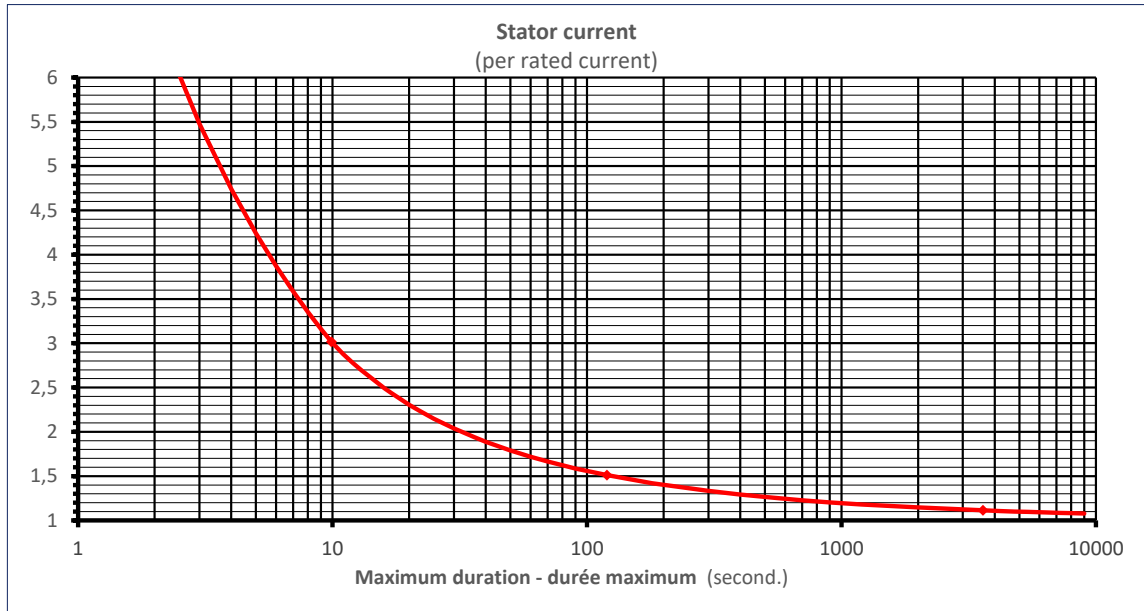


ALTERNATOR MAIN CURVES

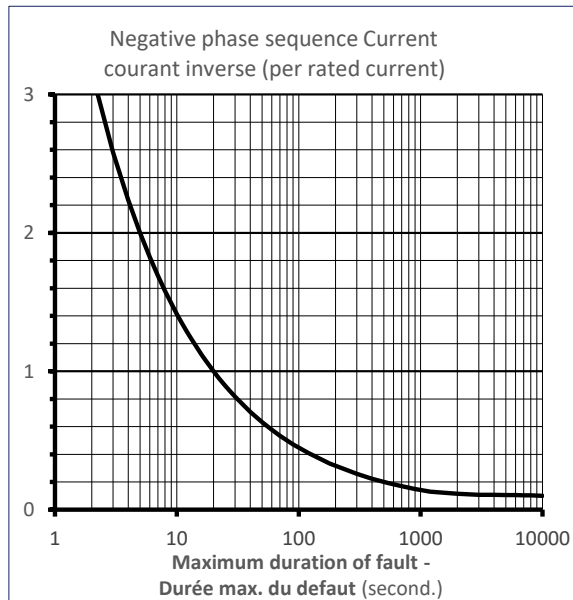
LSA 49.3 L9 / 4P

LS Reference: DS_A493_L9_415V_600ekW

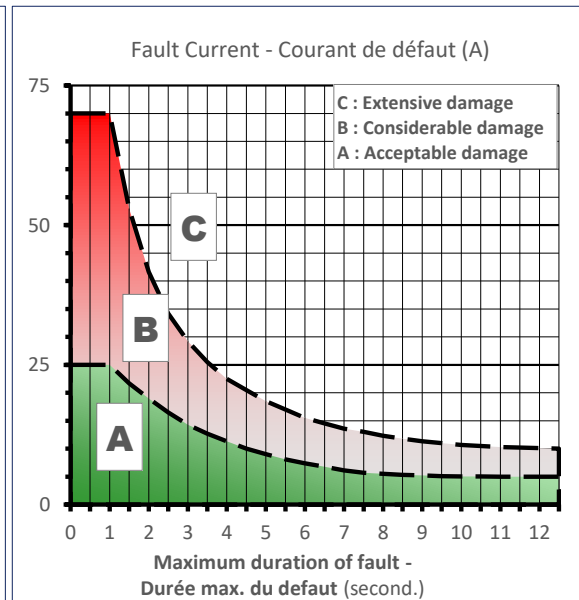
Thermal Damage Curve



Unbalance Load Curve



Stator Earth Fault Current



Foundation loading resulting from gas engine genset

Genset-drawing:	12528913	Base frame with oil tank
Technical information engine		

Engine:	CG132B-12	
Nominal frequency:	f_N [Hz]	50
Output:	P [kWel]	600
Speed:	n [min ⁻¹]	1500

Technical information alternator

Alternator:	Typ:	Drawing:
Voltage:	U [V]	400 - 415
Speed:	n [min ⁻¹]	1500
Short circuit moment:	M_k [kNm]	126,4

Technical Information Elastic Mounting

Elastic mounting (el. Iso.):	Standard	Motor	Generator
		Part-Number: 1252 0859 KZ	1252 0859 KZ

Foundation load due to weight of unit

Total genset mass:	m_{ges} [kg]	7850
Static foundation load:	F_{stat} [kN]	77,0
Degree of insulation of elastic mounting:	η [%]	94,3
Natural frequency of the elastic mounting:	f [Hz]	6

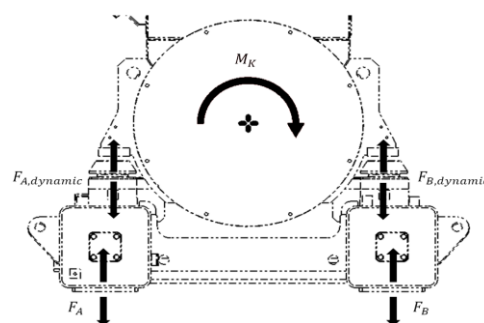
Foundation load in the event of a 2-phase alternator short circuit

This load will be exerted reciprocally irrespective of the direction of rotation

Maximum dynamic mounting-load at alternator short circuit (A-Side)	F_{dyn_A} [kN]	-10
Maximum dynamic mounting-load at alternator short circuit (B-Side)	F_{dyn_B} [kN]	10

Foundation load (static + dynamic)

A-Side		
Total load of genset	F_A [kN]	29
B-Side		
Total load of genset	F_B [kN]	48



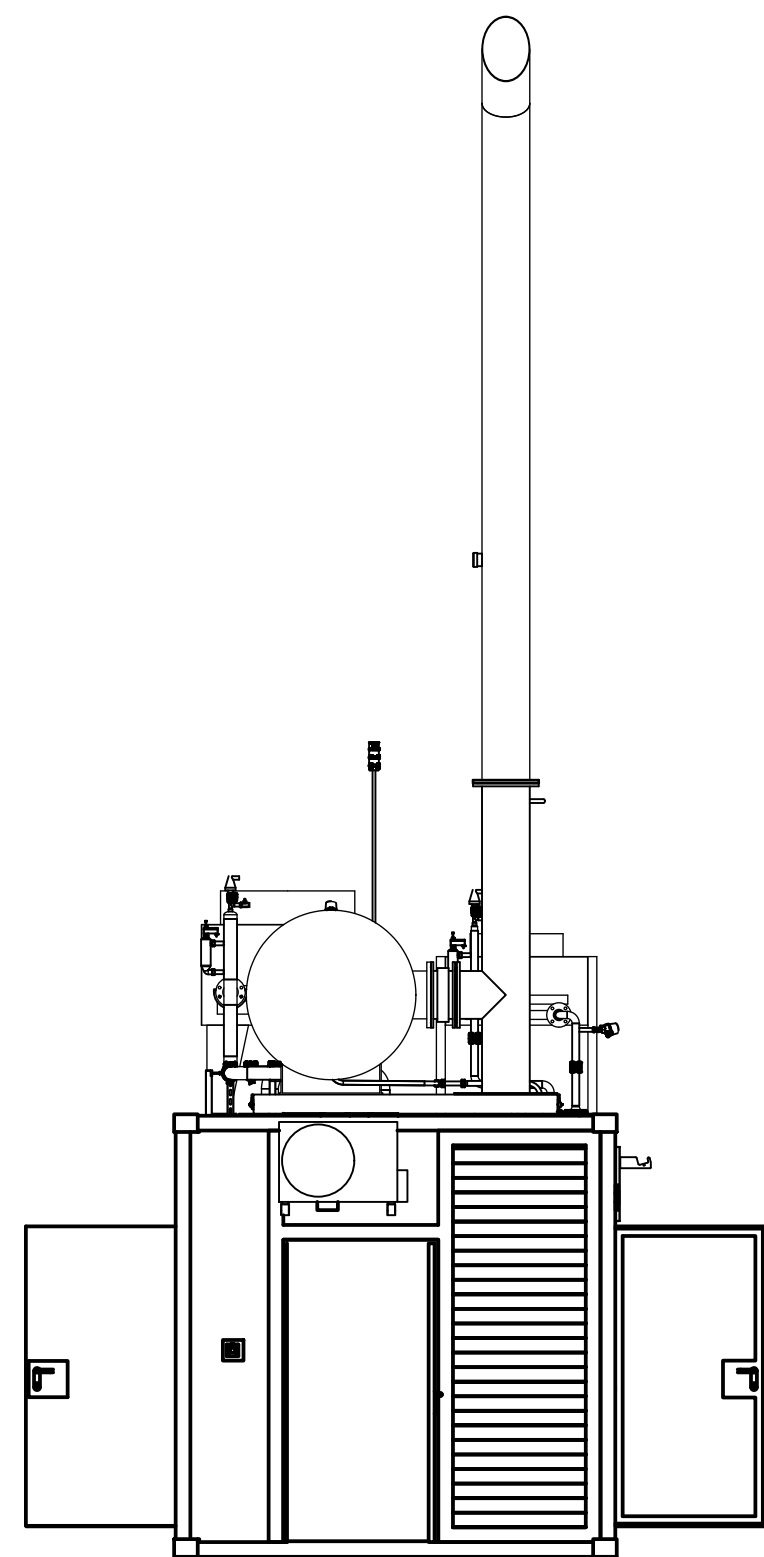
Condition of foundation surface referred "POWER PLANTS LAYOUT"

Datum:	05.11.2020	Create	Proved
		D. Gasper	U. Selzer

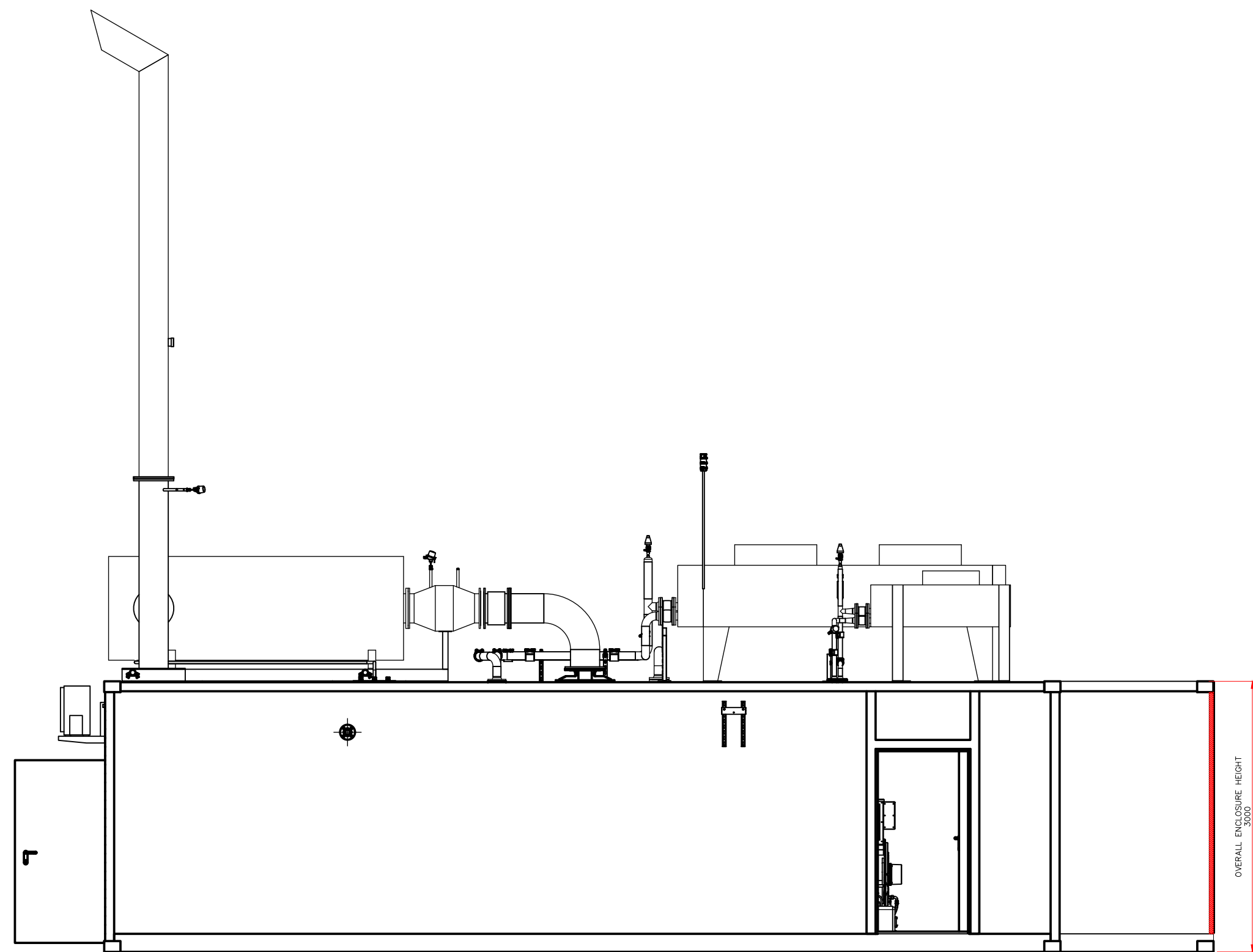
MR FOR GAS POWER GENERATOR							
SL. NO.	DESCRIPTION	UNIT	PROJECT REQUIREMENTS	VENDOR COMPLIANCE (CONFIRMED / NOT CONFIRMED)	CONTRACTOR REMARKS	CLIENT REMARKS	REMARKS
1.0	GENERAL						
1.1	Generator Rating	-	525kW, 415V, 3Phase, 4Wire, 50Hz, Containerized type Gas Engine Generator - 2 Nos. Star Connection, Duty S1	596ekW site rating at 45degC. When Ambient > 45degC Mist cooling in remote radiator will minimize deration with minimum water consumption			
1.2	Temperature	5-58 deg C	Vendor to confirm compliance				
1.3	Design Life	-	05 years; Bidder to comply	As per Cat maintenance schedule			
1.4	Vendor Deviation Form as per Attachment-6	-	Signed / Stamped copy is required with Offer Any deviations from the PROJECT requirements shall be listed in Attachment-6. Deviations listed elsewhere will not be considered for evaluation				
1.5	Tag No's		G-4101-01A & G-4101-01B				
1.6	Weight	kg	Vendor to Furnish	Dry weight - 7850kg			
1.7	Compliance to IEC 60034	-	Required				
1.8	Vendor & Sub-Vendors (Vendor of all items in scope) to be as per COMPANY Approved Vendor List	As per the Attachment-5 Approved Vendor List	Vendor to Confirm.	Complied			
1.9	Hazardous Area Classification	-	Safe Area	Complied , all measures in accordance with ISO 12100			
1.10	Anchor Bolt	-	Required				
1.11	Manufacturer	-	Vendor to Furnish	Catertpillar			
1.12	Model No.	-	Vendor to Furnish	CG132B-12			
1.13	Country of Origin	-	Vendor to Furnish	Germany			
1.14	Foundation Load Data	-	Vendor to Furnish	Attached : "Foundation load data"			
1.15	Installation/fixing detail with Foundation	-	Vendor to Furnish	As per attached genset package enclosure drawing			
1.16	Documents Required with Offer		Filled and Complied This MR Sheet Detailed Vendor Standard Datasheet of all components Vendor & Sub-Vendor List with port of origin Heat & Material Balance Diagram PID Equipment List Power Auxiliary List Consumption List Equipment Vendor Standard Catalogs MTC and Certificates Once technical closure is recieved signed stamped this MR Sheet	Attached supporing docs include: 1. PID 2. Equipment list 3. Power Aux list 4. Consumption list 5. Certification and standards			
2.0	Technical Requirements & Scope of Supply						
2.1	Alternator	-	Vendor to Furnish	Leroy Sommer			
2.2	Acoustic Enclosure with complete package skid	-	Required with cooling fans Enclosure shall be strong enough to withstand the CO2 release and corresponding CO2 thermal expansion	Complied			
2.3	Starting System	-	Vendor to Furnish	Electric starting			
2.4	Lube Oil System	-	Vendor to Furnish	As per PID			
2.5	Generator Duty Type	-	S1 according to IEC 60034-1	Complied			
2.6	Generator Insulation Temperature	-	Class F and to Class B temperature rise	Complied			
2.7	Generator Winding Insulation	-	Class H	Complied			
2.8	Installation requirement	-	Outdoor Installation in Containerized	Complied			
2.9	Fire & Gas System	-	F&G Detector Requirement 1. Flammable gas detection on the combustion and ventilation air inlet- Minimum 3 quantity 2. Gas detection on the ventilation air inlet -Minimum 3 quantity 3. Flame Detector at Enclosure including Load Compartment 4. Rise of Heat Detector at Enclosure including Load Compartment 5. Fire and Gas Beacon alarm & Horn - outside Enclosure Detector design The F&G devices shall be of minimum SIL 2 certified	UV adn IR sensors/ Detectors & fire extinguishers will be provided (beacons Alarms and Horns included)			
2.10	Fire Protection System	-	Required (automatic fire protection system using CO2 as the extinguishing medium	Complied			
2.11	Enclosure Temperature monitoring system	-	Required	Complied , Air conditioned electrical enclosure section			
2.12	Sustained Overload	-	Generator shall be capable of carrying a stator current of 110% of the rated current for a period of at least one hour at rated voltage, frequency and power factor.	Continuous rating : Output available with non-varying load for an unlimited time. Average power output is 70-100% of the continuous power rating. Typical peak demand is 100% of continuous rated ekW for 100% of operating hours			
2.13	AVR	-	Required (microprocessor based)	5%			
2.14	Voltage Variation	%	±10%				
2.15	Frequency Variation	%	±5%	0.4% regulation isynchronous			
2.16	Power Factor	-	0.8 lagging	Complied			
2.17	I ² t	-	Vendor to Furnish	25kw			
2.18	Electromagnetic Compatibility	-	Accordance to IEC 61000	According to : I.E.C. 60034.1 - 60034.2 - NEMA MG 1-32			
2.19	Starting Batteries	-	Required	Complied			
2.20	Type of Batteries	-	As per Manufacturer standard	Complied			
2.21	Battery Charger	-	Required	Complied			
2.22	Rating Plate	-	Required	Complied			
2.23	Efficiency (50 %, 75 %, and 100 % rated output)	-	Vendor to Furnish	25% 50% 75% 100% 110%			
2.24	Generator Cooling	-	Air Cooling or Liquid Cooling (Air cooling is preferred) Vendor shall include the number of fans and wattage	Power factor - Lagging: 0.8 92.90% 95.41% 95.90% 95.88% 95.81% Air cooled			
2.25	Generator Housing	-	IP 55 with TEFC enclosure (containerized)	IP-23 Alternator with Genset Enclosure IP 4x			
2.26	Neutral Earthing	-	Solidly Earthed for LV System	Complied			
2.27	Direction of rotation	-	IEC 60034-8	Complied			
2.28	RTD	-	PT100 RTD, 2 per phase (RTDs shall be connected to the DCS)	Complied			
2.29	Terminal Boxes	-	Separate terminal boxes shall be provided for: 1. Stator connections and Stator star point connections 2. Heater connections 3. Exciter connections 4. RTD connections 5. CT connection 6. Other connections	Complied (Top cable entry and breaker Termination for customer)			
2.30	Frequency stability	-	Governing class A2	G2			
2.31	Gas engine	-	Vendor to Furnish	Complied			
2.32	Air Intake System with filter and DPI and Wild-run shutdown valve	-	Vendor to Furnish (shall be part of the skid package)	Complied with G3 filtration & gravity shutters when not operational			
2.33	Engine Exhaust System with Silencer	-	Vendor to Furnish	Complied			
2.34	Terminal Boxes	-	Vendor to Furnish	Complied			
2.35	Engine Cooling System	-	Vendor to Furnish	Complied as per PID			
2.36	Shaft Couplings	-	Required	Complied , Included in package			
2.37	Torsional Analysis of Engine	-	Required	Complied			
2.38	Excitation system	-	Brushless type, consisting of a main AC Exciter and a permanent magnet generator direct coupled and air cooled	Complied			
2.39	Excitation - Rated Current	-	At least 110% of the excitation current at the rated output of the generator under the most stringent operating conditions				
2.40	Excitation - Rated Voltage	-	At least 110% of the rotor voltage at the rated output of the generator under the most stringent conditions				
2.41	Replacement interval of the lubricating oil	-	Vendor to Furnish	Maintenance schedule attached			
2.42	Noise Limit	dBA	85dB(A) at 1 meter distance	Complied			

MR FOR GAS POWER GENERATOR							
SL. NO.	DESCRIPTION	UNIT	PROJECT REQUIREMENTS	VENDOR COMPLIANCE (CONFIRMED / NOT CONFIRMED)	CONTRACTOR REMARKS	CLIENT REMARKS	REMARKS
2.43	Auxiliary Load Requirement	-	Vendor to Furnish	Included in the offer			
2.44	Anti-condensation Heaters	-	Required	Complied			
2.45	Engine and Generator Painting	-	As per Manufacturer standard	As per Cat standards			
2.46	Generators Synchronization	-	Generator synchronization relays shall be provided by Generator vendor as per attached Synchronization Scheme	Complied as per SLD			
2.47	Gas Pressure Regulating Skid	-	Generator synchronization relays shall be provided by Generator vendor as per attached Synchronization Scheme	Complied as per SLD			
2.48	The generator protection system and trip logic shall be in line with IEEE guidelines IEEE C37.101 and IEEE C37.102.	-	Vendor to confirm compliance	Complied			
2.49	Internal and External Earthing	-	Container shall be provided with internal earthing with suitably designed earth bars alongwith earthing cables, point for generator star earthing, external enclosure earthing ensuring safe and reliable operation	Within enclosure package along with accessories earthing is included			
2.50	Internal Lighting	-	Container shall be provided with internal lighting and external bulk head lighting on the doors	Complied			
2.51	Selective Catalytic Reduction system (SCR)	Optional Price	Vendor shall include the price based on the overall design and compliance to the project emission requirement	Complied			
2.52	Gas Genset Auxiliary Load	Refer to the attached Load List. Vendor shall review the load list and confirm the starting of the largest motor at star-delta configuration	Vendor to confirm compliance	Complied with transient as per G2 standards			
2.5	Gas Composition Data	Attachment-1A Fuel Gas_Sus_Summer Attachment-1B Fuel Gas_Sus_Winter	Vendor to confirm compliance	Complied operational with associated gas provided			
2.5.1	H2S Content	700 ppm	Vendor to confirm compliance	Complied . Active coal filtration included			
2.6	Inlet Gas Supply						
2.6.1	Pressure	Attachment-1A Fuel Gas_Sus_Summer Attachment-1B Fuel Gas_Sus_Winter	Vendor to confirm compliance	Complied operational with associated gas provided			
2.6.2	Temperature	Attachment-1A Fuel Gas_Sus_Summer Attachment-1B Fuel Gas_Sus_Winter	Vendor to confirm compliance	Complied operational with associated gas provided			
2.7	Ambient requirements			When Ambient			
2.7.1	For Gas Engine generator set	4-58 deg C	Vendor to confirm compliance	> 45degC Mist cooling in remote radiator will			
2.8	Gas type			minimize deration with minimum water consumption			
2.8.1	Biogas or Natural Gas	Dry Gas	Vendor to confirm compliance	Dry dehydrated gas only			
2.8.2	Emission Data	Compliance with Emission limits as per local regulation. Refer Attachment-2 Emission Compliance Data	Vendor to confirm compliance	Will be provided			
2.8.3	Emission Control Equipment	As required	Vendor to confirm as per the Emission Data requirement	SCR catalytic converter with oxidation catalytic converter			
2.9	Applicable Codes and Standards	Attachment-4 Applicable Codes and Standards	Vendor to confirm compliance	https://caterpillar-energy-solutions.de/en/success-factors/quality-management/			
3.0	Control and Protection Requirement						
3.1	Local Engine Control Panel (LCP) Placed inside the Gas Generator Container	-	Required, 1. Protection, control and metering equipment 2. AVR 3. Complete AC sub-distribution for the generator space heater, panel space heater and any other generator auxiliary items 4. Raise/lower voltage adjuster (behind the panel door) 5. Generator status, alarm and trip functions 6. Automatic starting / synchronizing facilities with manual override, synchroscope and synchronizing lamps 7. Control of switchgear remote generator CB 8. Control gear for generator set auxiliaries 9. Control relays (under-voltage, etc...) 10. Battery backed up DC power supply distribution for the generator protection systems (may be combined with the engine protection systems) 11. Generator testing facility 12. Winding/bearing temperature monitoring facility Vendor shall provide Redundant serial communication (RS485) for interface from Gas Generator Vendor provided LCP to Control system (UCP). 13. Fire Alarm & Detection and Fire Fighting System Equipment, Panel and Interface. 14. Gas Generator Enclosure external fans power and control panel supply with builtin contactors and external cooler temperature interface.	Complied with scope of supply included in offer			
3.2	Remote Generator Control Panel (GCP) Placed inside the Substation	-	Required (including control and monitoring equipment and engine measurements/controls) for all the components inside the Gas Generator Enclosure alongwith external fans through the HMI installed on the panel. Vendor shall furnish details, the panel shall be installed inside the substation. Vendor shall provide Redundant serial communication (RS485) for interface from Gas Generator Vendor provided LCP to Control system (UCP).	Will be provided included in GCP			
3.3	Protective Devices	-	All protective relays shall be of microprocessor type. The following electrical protective relays shall be provided as minimum to protect the generator. a. Reverse power (32) b. Stator winding over temperature (49) c. Time delay zero sequence over current (51N) d. Time delay voltage restrained over current (51V) e. Synchro check relay (25) f. Undervoltage relay (46) g. Negative phase sequence relay (46) h. Lockout relay (86) i. Over voltage protection (59) j. Field failure protection (loss of excitation) (40) k. Diode failure protection l. Trip circuit supervision relay (96) m. Instantaneous over current relay (50) n. Rotor earth fault (64R) o. Stator earth fault (64S) p. Frequency relay (81) q. Restricted Earth Fault Protection (87N) r. Differential Protection for Generator (87G)	Included in protection			

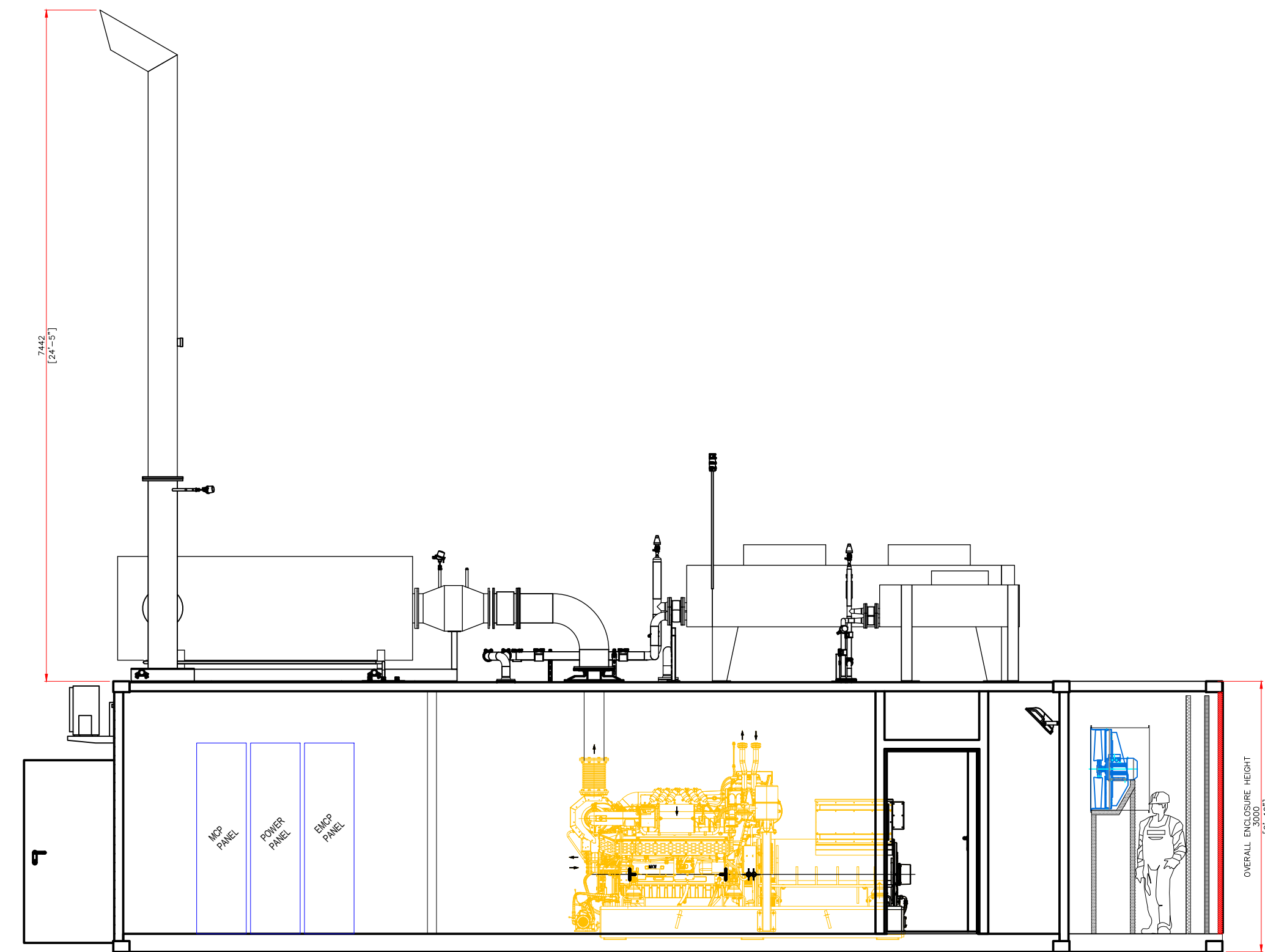
MR FOR GAS POWER GENERATOR							
SL. NO.	DESCRIPTION	UNIT	PROJECT REQUIREMENTS	VENDOR COMPLIANCE (CONFIRMED / NOT CONFIRMED)	CONTRACTOR REMARKS	CLIENT REMARKS	REMARKS
3.4	Interface with DCS & ESD		Gas Engine Driven Generators shall be interfaced to DCS and ESD system for monitoring, control and shutdown functions. Vendor shall provide hardwired and serial link interface signals for monitoring, control and shutdown functions. Interface with ESD shall be hardwired via Interposing Relay Panel MODBUS interfacing shall be utilized for the serial communication signals	HMI for monitoring will be provided with GCP			
3.5	On-line testing facility	-	Required for Generator				
3.6	Emergency Shut down facility including the shutdown valve and required control valves shall be provided as part of the package	-	Required	Complied			
3.7	Internal Wiring and Terminations	-	Vendor to provide	Complied			
3.8	Panel Painting	-	As per MANUFACTURER's standard	Complied as per standard			
3.9	Instrument Specification	-	As per MANUFACTURER's standard	As per standards			
3.10	Earthing	-	Vendor to provide a busbar connecting all the conducting material properly earthed with adequate earth cable on the skid baseplate.	As per standards			
3.11	Internal Tripping	-	Vendor shall consider internal Tripping signal and control signals within LCP.	Complied in generator control panel			
4.0	COMPLIANCE TO INTERNATIONAL STANDARDS						
4.1	IEC, BS & NFPA, NEMA	-	Required	Complied			
5.0	GENERAL REQUIREMENTS						
5.1	Quality and Vendor data Requirements	-	As per VDRL	As per Cat standards			
5.2	Painting	-	As stated above, Before shipment, all equipment's and components of the unit shall be Facility Painted to international standards, marine grade suitable for humid, dusty and saline site conditions,	As per Cat standards			
6.0	INSPECTION AND TESTING						
6.1	Insulation Resistance Tests	-	Required	Will be provided			
6.2	Hydrostatic Testing	-	Required	Will be provided			
6.3	Load Test	-	Required	Will be provided			
6.4	Dynamic Load Test	-	Required				
6.5	FAT	-	Required	Will be provided			
6.7	On Site Tests (SAT)	-	Required	Fuel as per Site for load test			
7.0	DOCUMENTATION						
7.1	Documentation	-	As per VDRL	Complied			
7.2	Language	-	English	Complied			
7.3	VENDOR Document Requirement - VDRL	Attachment-09 VDRL This list represents the minimum list of documents. Vendor shall include all required for manufacturing and document which are deemed necessary for quality control and commissioning	Vendor to confirm compliance	Complied			
8.0	SPARE PARTS & SPECIAL TOOLS						
8.1	Pre-commissioning/ Commissioning / Start up spares	-	Required / Vendor to provide, Start-up & Commissioning Spares : As recommended by Vendor with item wise quantity and price break up (To be ordered with the equipment). As a minimum shall include 200% gaskets & 10% bolting (minimum 2 nos. Bolts/4 nuts per size) shall be provided per gasketed internal & external joints (for nozzles with B/F).	Complied			
8.2	Two Years Operation Spares	-	Required - List in SPIR Form to be submitted	12000Hrs spare will be provided			
8.3	Special Tools and Tackles for Erection, Pre-commissioning, Commissioning & Operation (if applicable)	-	Vendor to Furnish	Complied			
8.4	A list of special tools necessary to dismantle the GEG	-	Vendor to Furnish				
9.0	OTHER REQUIREMENTS						
9.1	Guarantee / Warranty (Months)	-	12 Months after commissioning	Complied			
9.2	Training Manual & Installation Operation, Maintenance Manuals	-	Required	Complied			
9.3	Supervision for Commissioning at site	-	Per diem rates	As per AESL standard			
9.4	Delivery location	-	Construction Site, UAE	Complied			
9.5	Delivery Duration from Purchase Order placed date	weeks	VTA	42 Weeks			
9.6	Fabrication / Inspection & Test Location	-	Vendor to Furnish	Package inspection at AESL facility			
9.7	Local Agent Details	-	Vendor to Furnish				
9.8	Vendor's Engineering & Procurement Location	-	Vendor to Furnish	Pakistan			
9.9	Preservation	-	Preservation suitable for 6 months storage in marine, hot and humid saline atmosphere	Complied			
9.10	Vendor shall supply Gas Engine Generator (including accessories) in assembled condition. Bidder to confirm.	-	Required	Complied			
ABBREVIATIONS : VTA: Vendor to advise TBA:- To be advised VTC: Vendor to comply VTF: Vendor to furnish NA: Not Applicable MOC: Material of Construction							



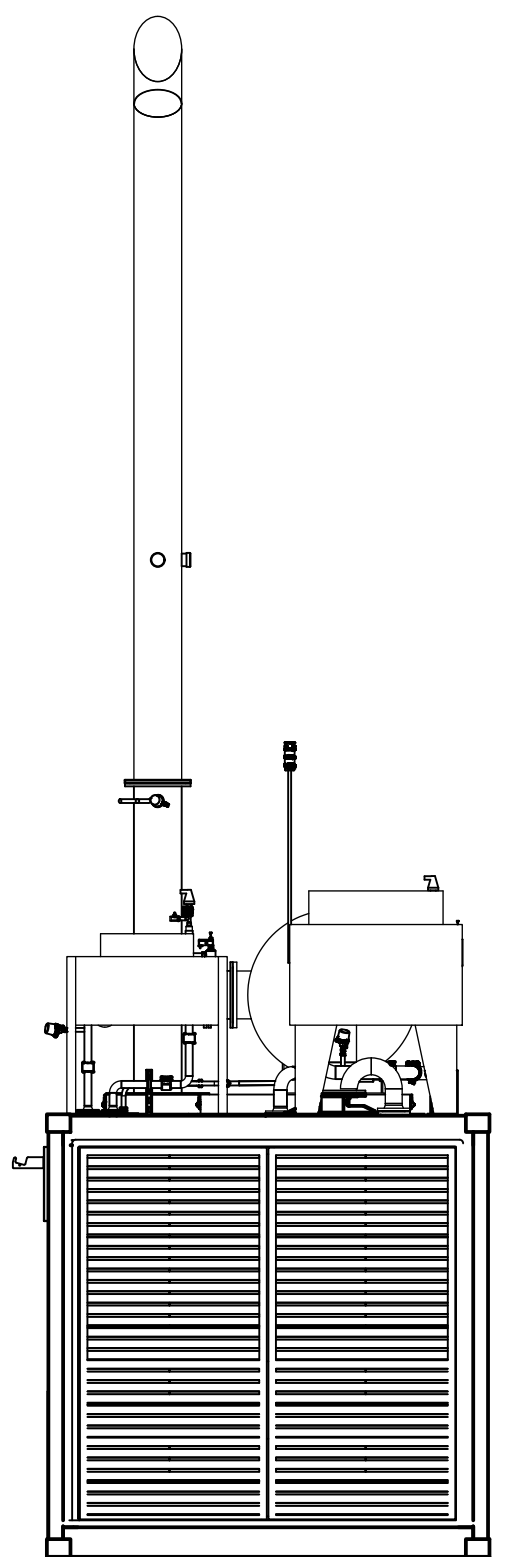
FRONT VIEW



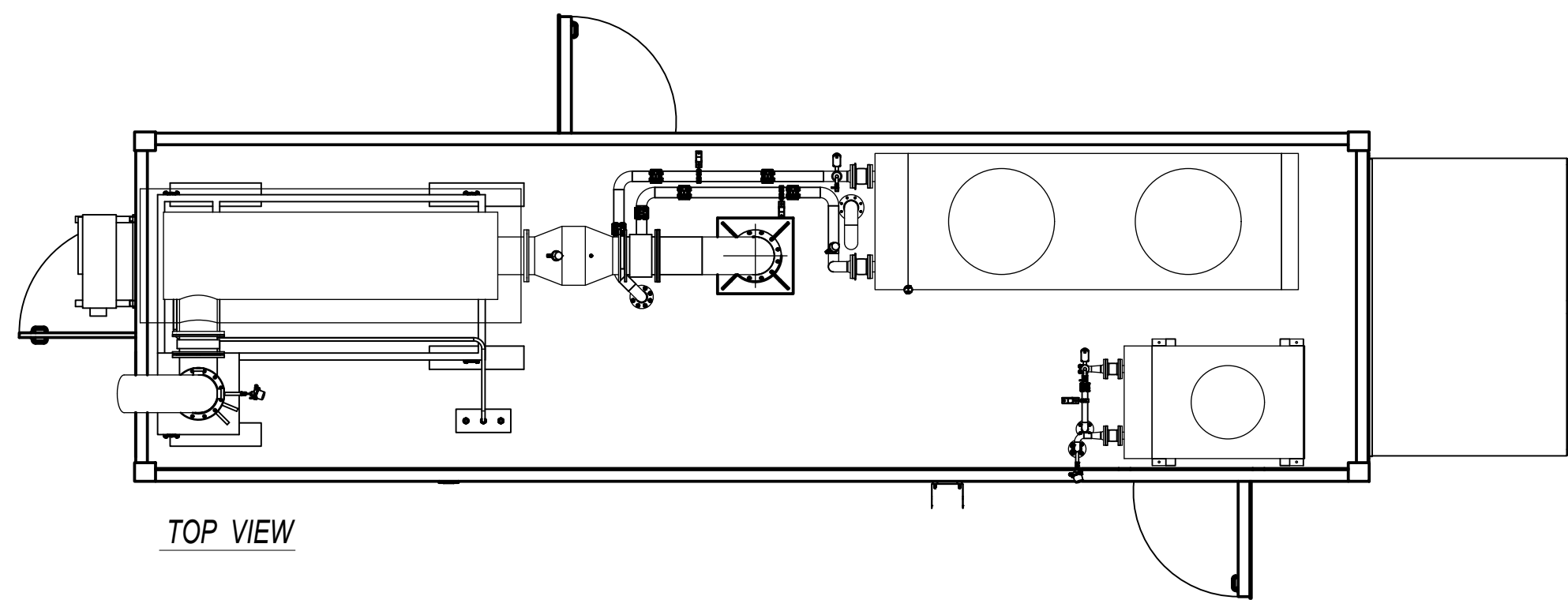
SIDE VIEW



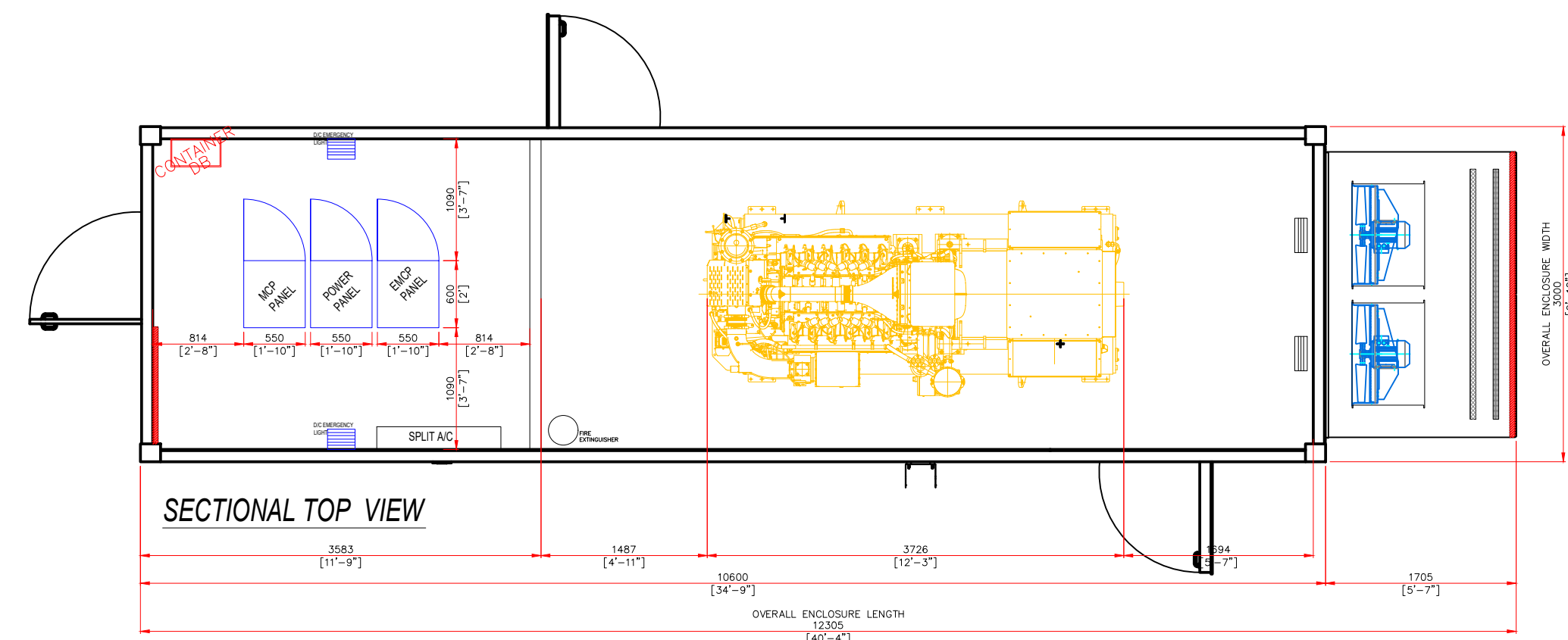
SECTIONAL SIDE VIEW



REAR VIEW



TOP VIEW



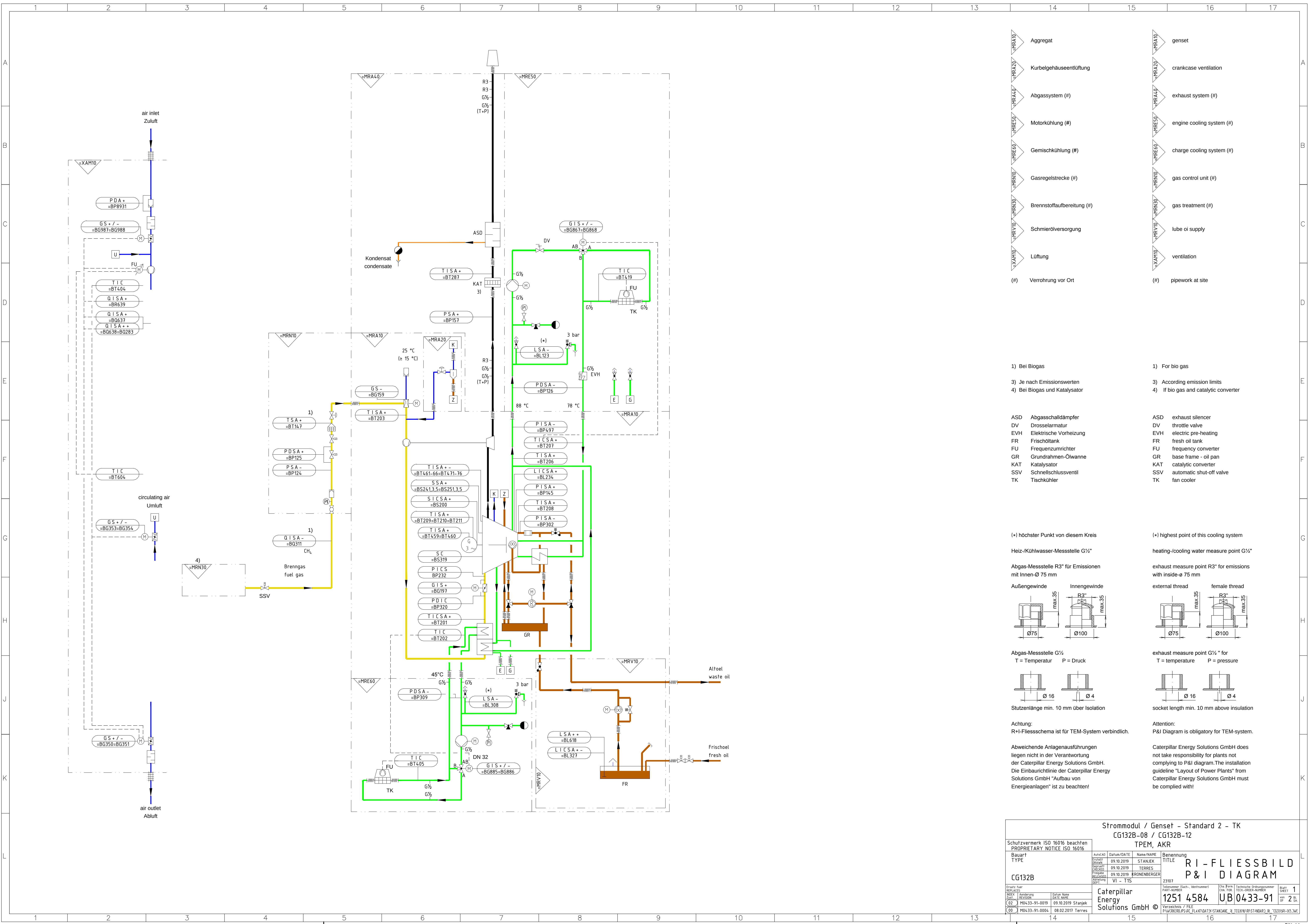
SECTIONAL TOP VIEW



REAR VIEW

SHEET 01 OF 01

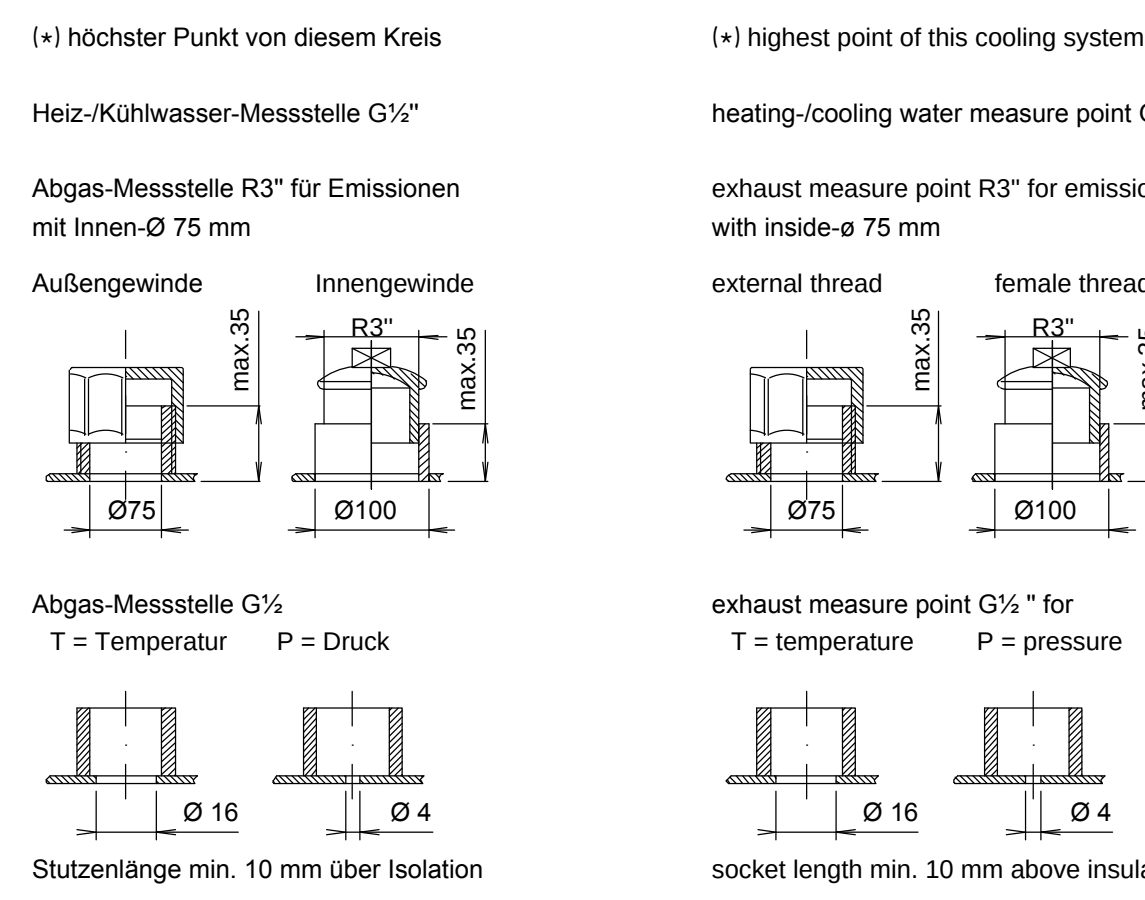
RevNo	Date	Revision note			
ALLIED ENGINEERING & SERVICES (Pvt.) LTD 21/3 Sector No.22 Korangi Industrial Area UAN: (021)111-250-250 website: www.aesl.com.pk		 THIRD ANGLE PROJECTION	Project: ADNOC - CAT CG132 ENCLOSURE		
			Title: CG 132 ENCLOSURE 600KW		
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM (INCH)		Model	AESL DRWG NO. AESL/CANOPY/CG132		Sheet 1/1 Scale NTS
Drawn by ATTA		Checked by SOFFYAN		Approved by	Filename OCT-24 Date 21-10-2024



- | | |
|----------------------------|---------------------------|
| Aggregat | genset |
| Kurbelgehäuseentlüftung | crankcase ventilation |
| Abgassystem (#) | exhaust system (#) |
| Motorkühlung (#) | engine cooling system (#) |
| Gemischkühlung (#) | charge cooling system (#) |
| Gasregelstrecke (#) | gas control unit (#) |
| Brennstoffaufbereitung (#) | gas treatment (#) |
| Schmierölversorgung | lube oil supply |
| Lüftung | ventilation |
| (#) Verrohrung vor Ort | (#) pipework at site |

- | | |
|-------------------------------|---------------------------------------|
| 1) Bei Biogas | 1) For bio gas |
| 3) Je nach Emissionswerten | 3) According emission limits |
| 4) Bei Biogas und Katalysator | 4) If bio gas and catalytic converter |

- | | | | |
|-----|------------------------|-----|--------------------------|
| ASD | Abgasschalldämpfer | ASD | exhaust silencer |
| DV | Drosselarmatur | DV | throttle valve |
| EVH | Elektrische Vorheizung | EVH | electric pre-heating |
| FR | Frischöltank | FR | fresh oil tank |
| FU | Frequenzumrichter | FU | frequency converter |
| GR | Grundrahmen-Ölwanne | GR | base frame - oil pan |
| KAT | Katalysator | KAT | catalytic converter |
| SSV | Schnellschlussventil | SSV | automatic shut-off valve |
| TK | Tischkühler | TK | fan cooler |



Achtung:
R+I-Fließschema ist für TEM-System verbindlich.

Abweichende Anlagenausführungen liegen nicht in der Verantwortung der Caterpillar Energy Solutions GmbH. Die Einbaurchlinie der Caterpillar Energy Solutions GmbH "Aufbau von Energieanlagen" ist zu beachten!

Attention:
P&I Diagram is obligatory for TEM-system.

Caterpillar Energy Solutions GmbH does not take responsibility for plants not complying to P&I diagram. The installation guideline "Layout of Power Plants" from Caterpillar Energy Solutions GmbH must be complied with!

Strommodul / Genset - Standard 2 - TK									
CG132B-08 / CG132B-12									
TPem, AKR									
Schutzvermerk ISO 16016 beachten PROPRIETARY NOTICE ISO 16016				Benennung RI - FLIESSBILD P & I DIAGRAM					
Bauart TYPE				23107					
CG132B				1251 4584 UB0433-91					
Caterpillar Energy Solutions GmbH ©				Verzeichnis / FILE P\WCRCR\PS\RC_PLANT\DATA\STANDARD_3_T\CG132B\STANDARD_3_T\CG132B-08-25-26					
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